



**CONTINUOUS LOWER ENERGY, EMISSIONS and NOISE (CLEEN)
PROGRAM (2026-2028) Phase IV
PARTICIPANT
SELECTION**

**SCREENING INFORMATION
REQUEST 693KA9-26-R-00006**

Date: March 30, 2026

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1 Project Title

Continuous Lower Energy, Emissions and Noise (CLEEN) Phase IV Program (2026-2028)
Program Participant Selection

2 Screening Information Request (SIR) No.

693KA9-26-R-00006

3 Agreement Type

49 U.S.C. 106(l)(6) and/or 106(m) Cooperative Agreement

This SIR is intended to result in the award of Cooperative Agreement(s) with the FAA under the agency's authority set forth in 49 U.S.C. 106(l) and (m) (hereinafter referred to as the "Agreement"). A sample Agreement is attached to this SIR and is incorporated herein by reference. While the FAA anticipates that the Agreement will be in substantially the same form as the attached Agreement, the FAA has sole and absolute discretion as to the form and substance of each Agreement entered into with successful Respondents, and the attached sample Agreement may accordingly be modified in form or substance as the FAA deems appropriate.

The Contracting Officer (CO) is the signature authority for executing Agreements. The Agreement establishes the responsibilities of the FAA and the Lead Participant. The Acquisition Management System (AMS) establishes policy and guidance for all aspects of lifecycle acquisition management for the FAA. Respondents may obtain information on the AMS via the Internet at <http://fast.faa.gov/>. Some AMS policies are referenced in this document as an established protocol.

4 Eligibility

Respondents must be located in the United States. In accordance with Executive Order 14005, Ensuring the Future is Made in All of America by All of America's Workers, and consistent with the "America First" initiative, the federal government is directed to maximize the procurement of goods and services from domestic sources to support American manufacturing, innovation, and job creation. To comply with those directives, Respondents to this SIR must be incorporated in the United States (U.S.), primarily operate within the United States, and manufacture their products substantially within the United States.

Respondents must follow 2 CFR § 200.322. If, to the greatest extent practicable under Section 200.322, a U.S. product or service will not work for the project, then respondents must provide justification for the use of a non-U.S. product or service.

5 Period Of Performance

The period of performance for each Agreement is expected to be up to 24 months. Respondents may propose a period of performance of fewer than 24 months.

6 Funding

The CLEEN Phase IV program includes a 1:1 (one-to-one) minimum cost share requirement. Contributions may be cash, non-cash (in-kind) or both in accordance with 2 CFR Part 200, Uniform Administrative Requirements, Cost Principles, and Audit Requirements for Federal Awards, including 2 CFR § 200.306, Cost sharing, and 2 CFR Part 200, Subpart E, Cost Principles.

Acceptable non-cash or in-kind resources include such items as equipment, facilities, labor, office space, etc. Unacceptable contributions include work performed using FAA or other U.S. funding, previous Internal Research and Development (IRAD) investments, student tuition funded by an academic institution, and unallowable costs under 2 CFR § 200. Participants must include cost share progress reports in the financial progress reports required under Section 8.6.

The FAA's total liability under the Agreement is no more than 50% of the total cost of certain work that will be specified in any Agreement awarded under this SIR. The Participant must cover at least 50% of the project costs. Participant proposed contributions may be more than the FAA would cost-share. In no event will the FAA's share be more than 50% of the total cost of the project specified in the Agreement or over the obligated amount of the Agreement. The FAA will not cost-share any costs or provide additional funding for projects beyond the obligated amount of the Agreement. The work will be related to the development of continuous lower energy, emissions, and noise technologies for civil airplanes.

"Contributions" as used in this SIR is the portion of the total project that is not paid for by the FAA. There are no waivers or exemptions for the contribution requirements.

General parameters regarding cost share as they relate to the CLEEN IV program include:

- Participants should first determine whether the intended Participant contribution is eligible under CLEEN IV. An "eligible expense" is a Participant expenditure that directly facilitates the execution of the Agreement and is "reasonable," "necessary," and "allocable" (2 CFR § 200.403[a], 404, and 405). Eligible expenses comprise the total project cost for the project. The FAA will provide cost share funds for up to 50% of the total project cost.
- Individual eligible expenses incurred by the Participant may either be reimbursed with FAA CLEEN IV funding at 50%, or alternatively, not reimbursed and may count 100% toward the contribution, so long as the total overall Agreement demonstrates Participant cost share of at least 50% of the total overall Agreement costs.
- Contributions may be spread out over the period of performance so long as at the completion of the Agreement, the Participant's 50% cost share requirement is met.
- The cost share from an individual Participant can be determined on an overall basis relative to all of the goals that the Agreement is working toward, across multiple technology projects, as applicable. For example, if a Participant is awarded multiple technology projects, the cost share can be less than 1:1 for one technology project as long as the cost share is greater than 1:1 for other technology projects such that the overall cost share under the agreement meets at least the 1:1 minimum requirement.
- Cost share may include cash (*e.g.*, non-FAA funds from the Participant). These funds must be utilized to complete activities outlined in the Agreement.
- Cost share may include the value of labor, facilities, equipment and utilities provided by the Participant that directly facilitates the efforts required under the Agreement and may be billed as direct or indirect costs (2 CFR § 200.412, 413, and 415). These funds must be utilized to complete activities outlined in the Agreement.
- An eligible cost share expense for CLEEN IV may not be eligible in all instances across all like Federal programs.
- An eligible expense to be counted toward cost share must be documented appropriately to be considered.

The amount of FAA funding under the Agreement and the portion of the specified work to be funded under the Agreement, either at any particular time or cumulatively under the Agreement, are at the sole discretion of the FAA in all instances. The FAA's liability at any particular time or cumulatively under the Agreement must not exceed the amount identified as obligated under the Agreement and will not exceed the FAA's total liability under the Agreement. The FAA, at its sole discretion, may obligate additional funding to support this work. All funding is subject to the availability of funds.

The resulting Cooperative Agreement is subject to 2 CFR Part 200, Uniform Administrative Requirements, Cost Principles, and Audit Requirements for Federal Awards, including 2 CFR § 200.306, Cost sharing, and 2 CFR Part 200, Subpart E, Cost Principles. In accordance with 2 CFR § 200.305 (b)(3), the payment method for the resulting Cooperative Agreement is Cost Reimbursement.

Pre-award costs, which are those incurred before the start date of a Federal award, are not permitted under the Agreement. Pre-award costs are defined as any costs incurred before the start date of the federal award in anticipation of the federal award. Any pre-award costs are at the Respondent's risk. The FAA will not approve or pay any pre-award costs.

7 Background

As captured in Section 47511 of Title 49 of the United States Code, the Federal Aviation Administration is authorized to enter into a cost-sharing cooperative agreement, using a competitive process, with institutions, entities, or consortiums to carry out a program for the development, maturation, and testing of certifiable CLEEN aircraft, engine technologies, and jet fuels for civil airplanes where the term "CLEEN aircraft and engine technology" means continuous lower energy, emissions, and noise aircraft and engine technology. Section 47511 also requires the establishment of performance objectives for the CLEEN Program in terms of the specific objectives to reduce fuel burn, emissions, and noise. In addition, in carrying out the program, the FAA may provide that not less than 2 of the cooperative agreements involve the participation of an entity that is a small business concern (as defined in Section 3 of the Small Business Act (15 U.S.C. 632)), provided that the submitted technology proposal of the entity meets, at a minimum, FAA Acquisition Management System requirements and requisite technology readiness levels for entry into the agreement, as determined by the Administrator.

The FAA intends to establish on a cost share, 1:1 minimum basis, agreements for activities to mature Continuous Lower Energy, Emissions and Noise (CLEEN) technologies for civil airplanes and inform the development of FAA airframe and engine environmental standards and environmental review of rulemaking and other actions. This program will enable and expedite the introduction of innovative airframe and engine technologies with improved performance and safety and enhance the technological and economic competitiveness of American civil aircraft.

A challenge to novel aircraft technology research and development is ensuring that technologies are certifiable. Manufacturers must demonstrate that technologies meet airworthiness certification requirements—including environmental standards—and the FAA must ensure that certification requirements and methods are in place for these technologies. The FAA regulates engine emissions,

airplane noise, and airplane fuel efficiency through 14 CFR Parts 34, 36, and 38, respectively. Partnership through CLEEN enables the effective and efficient development of environmental standards that expedite the safe introduction of next generation technologies. This collaboration may also facilitate environmental review by allowing for a better understanding of the environmental and operational characteristics of new and emerging entrants. Continued partnership is critical to ensuring the safe and timely introduction of advanced innovative aircraft technologies with improved performance. Focused investments will create good-paying jobs within the U.S. aviation manufacturing industry and maintain U.S. global competitiveness by enabling innovation in the U.S. aviation industry. These efforts will benefit the U.S. flying public by virtue of increased performance and safe deployment of innovative technologies.

The next phase of the CLEEN Program, referred to as Phase IV, is a follow-on program from the first three five-year phases of CLEEN. Phase IV of the CLEEN Program will continue the Program through calendar year 2028. The fourth phase of CLEEN, like the first three phases of CLEEN, is a public-private partnership in which projects are cost-shared between the FAA and industry.

The FAA anticipates multiple awards from this solicitation. The total funding available for all projects is 25 million dollars for the duration of the program..

Concepts should be submitted for the full period of performance, including their current Technology Readiness Level¹ (TRL) and Manufacturing Readiness Level² (MRL), and projected concluding TRL and MRL levels.. The FAA reserves the right to exercise option periods at its sole discretion and does not anticipate holding an additional procurement cycle for the option period.

Actual amounts of awards will be specified for each agreement prior to award and are at the sole discretion of the FAA.

¹ Technology Readiness Levels

- TRL 1: Basic principle observed and reported.
- TRL 2: Technology concept and/or application formulated (candidate selected)
- TRL 3: Analytical and experimental critical function, or characteristic proof of concept.
- TRL 4: Component and/or breadboard test in a laboratory environment.
- TRL 5: Component and/or breadboard validation in a relevant environment
- TRL 6: System/subsystem model or true dimensional test equipment validated in a relevant environment.
- TRL 7: System prototype demonstrated in flight environment.
- TRL 8: Actual system completed and “flight qualified” through test and demonstration.
- TRL 9: Actual system “flight proven” on operational flight

² Manufacturing Readiness Levels

- MRL 1: Basic Manufacturing Implications Identified
- MRL 2: Manufacturing Concepts Identified
- MRL 3: Manufacturing Proof of Concept Developed
- MRL 4: Capability to produce the technology in a laboratory environment
- MRL 5: Capability to produce prototype components in a production relevant environment
- MRL 6: Capability to produce a prototype system or subsystem in a production relevant environment
- MRL 7: Capability to produce systems, subsystems, or components in a production representative environment
- MRL 8: Pilot line capability demonstrated; Ready to begin Low Rate Initial Production
- MRL 9: Low rate production demonstrated; Capability in place to begin Full Rate Production.
- MRL 10: Full Rate Production demonstrated and lean production practices in place

8 Description of Effort, Deliverables, and Requirements

8.1 Introduction

Phase IV of the CLEEN Program is focused on activities that will inform the development of FAA airframe and engine environmental standards and environmental review of rulemaking and other actions in order to enable and expedite the safe introduction of new airframe and engine technologies with improved performance and safety. Sections 8.2 and 8.3 provide the performance objectives and example project areas for this phase of CLEEN. The remaining subsections describe the structure of the program.

8.2 Performance Objectives

As defined by 49 U.S. Code § 47511, the performance objectives of CLEEN are to reduce fuel burn, engine emissions, and noise through the development, maturation, and testing of certifiable technologies for civil airplanes. For CLEEN Phase IV, an additional objective is to collect information that can assist the FAA in developing more effective and timely environmental standards and conducting environmental reviews, thereby supporting a pathway to type certification for proposed CLEEN technologies. Three types of information may be provided in support of this objective: 1) information that can assist the FAA in setting appropriate levels for environmental standards, 2) information that can assist the FAA in improving certification procedures from an environmental perspective where the existing regulations and policy do not address or are not adequate for a given technology, and/or 3) information that can assist the FAA in conducting environmental reviews of rulemaking and other actions.

While CLEEN Phase IV does not have explicit goals for noise, fuel burn (in terms of energy use) and emission reductions, to be eligible for submission, technologies must provide a benefit in at least one of these environmental areas relative to current aviation products. The relevant metrics are provided below for each environmental area:

- Fuel efficiency: Aircraft technology that improves aircraft fuel efficiency relative to the International Civil Aviation Organization (ICAO) new type fuel efficiency standard for civil subsonic airplanes adopted in 2016 (Committee on Aviation Environmental Protection, CAEP/10);
- Noise: Aircraft technology that reduces noise levels relative to the CAEP/13 Chapter 16 standard for civil subsonic airplanes and/or reduces local noise exposure;
- NO_x emissions: Engine technology that reduces landing and takeoff cycle (LTO) nitrogen oxide emissions relative to the ICAO standard for civil subsonic airplanes adopted in 2010 (CAEP/8) and/or reduces absolute NO_x production over the aircraft's mission. This must be achieved while limiting or reducing other gaseous or particulate matter emissions; and
- Particulate matter: Engine technologies that reduce LTO particulate matter number and mass emissions relative to the ICAO new type standard for civil subsonic airplanes adopted in 2019 (CAEP/11) and/or reduces absolute nvPM production over the aircraft's mission. This must be achieved while limiting or reducing other gaseous or particulate matter emissions.

Any technologies targeted at civil supersonic airplanes should demonstrate quantifiable benefits for fuel efficiency, noise, NO_x, and/or particulate matter emissions relative to current state of the art technology and design concepts.

Below are additional clarifying points on the performance objectives for the fourth phase of the CLEEN Program:

- CLEEN Phase IV does not include specific requirements for starting and ending TRL. Candidate technologies must be matured and demonstrated at a TRL sufficient to provide information to the FAA that supports the CLEEN IV performance objectives.
- Fuel efficiency improvement is in terms of whatever energy source the aircraft and propulsion system will be using and is equivalent to fuel efficiency for jet fuel powered aircraft. This goal may also include technologies that improve efficiency and/or reduce aviation's impacts in ways not captured by the fuel efficiency standard metric (*e.g.*, contrail understanding and mitigation).
- Absolute NO_x production over the aircraft's mission is to be measured relative to a 2020 best-in-class in-service baseline aircraft and mission.
- Both the nvPM number and mass need to be measured with an ICAO Annex 16 / SAE compliant measurement system.
- An aircraft technology or an operational/airspace integration capability that would enable an operational procedure concept that facilitates the above performance objectives is within the scope of Phase IV of CLEEN.

8.3 Example Project Areas

CLEEN Phase IV is not limited to specific technologies or aircraft types. Project proposals must support the performance objectives detailed in Section 8.2. However, there are emerging technology areas where there are potential gaps in knowledge or procedures related to environmental standards and environmental review. Below are examples of innovative technologies where the FAA could benefit from collaboration with industry to help inform the development of appropriate environmental standards, ensure certification readiness, and enable environmental review. This list is not exhaustive and should not be construed as indicative of FAA's project preferences when it comes to proposal evaluation.

- Non-traditional propulsion systems – *e.g.*, open rotor, dual fuel, electric/hybrid-electric, cryogenic fuel powered
- Advanced aircraft configurations – *e.g.*, blended-wing body, truss-braced wing, over-wing nacelle
- Supersonic aircraft and engines – *e.g.*, variable noise reduction systems (including potential application to subsonic aircraft), systems to enable Mach cut-off operations
- Advanced air mobility vehicles and unmanned aircraft systems

As noted in Section 8.2, the FAA is also interested in information that can assist the FAA in setting appropriate levels for environmental standards, including information for technologies for which the certification requirements and procedures are more well-established. The U.S. participates in the International Civil Aviation Organization (ICAO) Committee on Aviation Environmental Protection (CAEP) to develop and harmonize aircraft and engine environmental standards. As part of the standards setting process, CAEP conducts analyses of costs and benefits of different options for potential regulatory limits to ensure that standards are technically feasible, economically reasonable, and environmentally beneficial. Projects that can provide information that would assist the FAA in this standards setting process (*e.g.*, noise/emissions data, technology responses to different stringency options, etc.) may be supported under CLEEN

Phase IV.

Regarding environmental review, projects in the new and emerging aircraft entrants area may be most relevant. The FAA considers the impact of its actions through coordinated environmental review under the National Environmental Policy Act (NEPA) and other federal laws.

Understanding the environmental performance characteristics of these vehicles, as well as how and where they will operate, can help the FAA conduct the modeling and analysis activities necessary for environmental reviews of FAA actions that will enable emerging aircraft operations.

8.4 Aircraft Technology Work Elements

The aircraft technology work consists of five major Work Elements: (1) project management; (2) design; (3) fabrication, (4) procurement and (5) assembly; technology demonstration; and technology assessments and reporting. In these efforts, the Participant must establish the technologies and technology validation plans, design and develop necessary hardware, and perform tests that will constitute technology demonstrations, as applicable. Respondents may submit proposals that complete only a subset of Work Elements 2, 3, and 4 (*i.e.*, one or more of these work elements). All proposals must include Work Elements 1 and 5.

8.4.1 Aircraft Technology Work Element 1: Project Management

The Participant must conduct the program management activities needed to supervise, direct, and manage the demonstration of technologies pursuant to the performance objectives identified in Section 8.2. The Participant must employ those project management and planning tools considered necessary to ensure the accomplishment of each work element in a timely manner and within budget constraints. This Work Element includes the planning, monitoring, analysis and reporting of participant cost, schedule and technical performance.

Within 30 days of agreement award, the Participant must submit a Task Plan for FAA approval. The Task Plan must incorporate negotiated changes to the Technical Proposal, if any. The Participant must maintain and control the Task Plan during the execution of this task.

8.4.2 Aircraft Technology Work Element 2: Design

The Participant must accomplish the following activities for each proposed technology maturation activity, as applicable:

- The Participant must conduct all efforts necessary to design, develop, and demonstrate selected candidate technologies specified in the Task Plan. This effort includes requirements definition and design of prototypes, test articles and any test equipment required to support the technology development effort.
- The Participant must conduct the necessary requirements gathering and definition activities to ensure that the technology design will proceed to meet the performance objectives of CLEEN Phase IV.
- The Participant must conduct the necessary design reviews to demonstrate that system requirements are met and the maturity of the design is appropriate to support proceeding with fabrication, assembly, integration, and test, as applicable. The Participant must submit Design Review Packages to FAA that include:
 1. The mechanical, aerodynamic, electrical, and/or control system/logic design (as

- applicable) of the components and any testbed structure modifications that are required to incorporate these components into the testbed system.
2. A systems analysis of the testbed system that defines key operating characteristics and performance parameters of the testbed.
 3. An updated estimate of the anticipated benefit of each technology to be tested, in terms of reduction in noise, emissions, and/or fuel burn/energy use.
 4. A summary of consideration given in the definition of the design to manufacturability, maintainability, affordability, repairability and eventual certifiability of the technology.
- Throughout the design process, the Participant must employ those systems engineering and integration practices and tools considered necessary to ensure the accomplishment of the design to meet schedule, cost, and technical requirements.

8.4.3 Aircraft Technology Work Element 3: Fabrication, Procurement, and Assembly

The Participant must conduct the necessary fabrication, procurement, and assembly activities to complete the test articles, test equipment and special tools required to support the technology development and demonstration effort, as applicable.

The Participant must conduct manufacturing trials, as needed, to verify and/or advance the manufacturing readiness level of the technology.

Throughout the fabrication, procurement, and assembly process, the Participant must employ those systems engineering and integration practices and tools considered necessary to ensure the accomplishment of these efforts to meet schedule, cost, and technical requirements.

8.4.4 Aircraft Technology Work Element 4: Technology Demonstration

This activity must include the tests required to demonstrate the technology, as applicable.

The Participant must conduct rig, ground, and/or flight tests for the baseline aircraft and/or engine, and modified aircraft and/or engine configuration to demonstrate the proposed technologies. The Participant must establish baseline and advanced technology modified configuration measurements for use in evaluation of achievement of the CLEEN Phase IV performance objectives. These measurements must include, as applicable:

- Data necessary to inform the FAA in developing environmental standards and/or conducting environmental reviews.
- Data necessary to evaluate and quantify the fuel burn/energy use benefit of related technologies.
- Engine and/or airframe noise measurements for use in the evaluation of the noise reduction technologies, including detailed explanations of the spectral features of those measurements.
- Data necessary to evaluate and quantify the NO_x emissions benefit of related technologies.
- Gaseous and non-volatile particulate matter (nvPM) emissions measurements in the baseline engine and the advanced technology configuration. Non-volatile particulate matter emissions measurements must be obtained with an ICAO Annex 16 Vol. II Appendix 7/SAE ARP6320 compliant measurement systems. System loss calculation of

both nvPM mass and number emissions must be calculated and reported using ICAO Annex 16 Vol. II Appendix 8/SAE ARP6481. In preparation for ARP6320 and 6481 revisions, size distribution measurements must be incorporated per SAE AIR7382 and AIR6504A.

Note that in the case of a flight management system (FMS) or air traffic management (ATM) related technology, these demonstration tests may be achieved by modeling and simulation in the relevant environment. Such demonstrations must also be conducted in such a manner so as to gather the data necessary to inform FAA environmental standards development and environmental reviews and to evaluate the benefits of a given technology with respect to the CLEEN Phase IV performance objectives.

Throughout the technology demonstration process the Participant must employ those systems engineering and integration practices and tools considered necessary to ensure the accomplishment of technology demonstration efforts to meet schedule, cost, and technical requirements.

The Participant must submit the following Deliverables for technology maturation activity under this Work Element, as applicable:

- A demonstration plan in support of Section 8.2 performance objectives. Such plan must be submitted to the FAA Task Manager for review and approval by the date established by the FAA, in its sole discretion, on a project by project basis, but in any event in advance of the initiation of all proposed test demonstrations. This documentation must include information on test objectives, instrumentation, test facility and setup, and test points to be run.
- An informal test report submitted to the FAA Task Manager for review and approval by the date established by the FAA, in its sole discretion, on a project by project basis. The report must include information on the test objectives accomplished and data acquired.
- A final test report submitted to the FAA Task Manager for review and approval by the date established by the FAA, in its sole discretion, on a project by project basis. This report must include information on the test objectives accomplished, data acquired, interpretation and value of the data, and an updated assessment of the anticipated technology benefit, in terms of reduction in noise, emissions and/or fuel burn/energy use.
- The Participant must provide a final presentation to the FAA Task Manager two weeks after submitting the final test report. The presentation must include information on the test objectives accomplished, data acquired, interpretation and value of the data, and an updated assessment of the anticipated technology benefit in terms of reduction in noise, emissions and/or fuel burn/energy use.

8.4.5 Aircraft Technology Work Element 5: Technology Assessments and Reporting

The Participant must quantify the benefits of the developed technologies in terms of noise, emissions, and/or fuel burn/energy use. The Participant must quantify any effects of the technology on aircraft and/or engine performance, as well as any tradeoffs across CLEEN IV performance objectives. The Participant must identify any issues that would impact manufacturability, maintainability, affordability, and/or repairability and/or prevent certification of the technology to existing noise, emissions, and airworthiness standards. The Participant must provide a timeline and key activities to further technology maturation after the conclusion of the

fourth phase of the CLEEN Program.

- The Participant must provide the necessary component level characteristics, operating limits, and input data that will enable system-level modeling of the proposed technologies by the U.S. FAA. The Participant must provide all necessary model verification data to the FAA Task Manager as the FAA CLEEN team will verify its technology modeling and associated benefits.
- The Participant must allow an independent third party, designated by the FAA in its sole discretion, to conduct the benefits assessments and to share all findings with the FAA without the Participant's approval. The data provided will also be utilized by the U.S. FAA or U.S. FAA contractors under a properly executed non-disclosure agreement to the extent necessary to perform the system-level modeling tasks. The FAA will determine the specific data that the Participant must provide based on the system-level modeling data requirements for the technology that is proposed for demonstration. This information will be used for internal U.S. FAA assessment of the advanced concepts, will be restricted to U.S. FAA use, and will be treated as procurement sensitive information.

8.5 Intellectual Property

The FAA intends to negotiate intellectual property rights with each Participant. The following paragraph expresses the FAA's overall intentions as to intellectual property, but such goals are subject to change in the course of negotiations with each successful Participant.

The FAA plans to utilize data acquired through this research to advance the development of FAA airframe and engine environmental standards and conduct environmental reviews. The FAA plans to share the details of this research as required to advance the program goals within the U.S. Aerospace community in a timely manner. The FAA intends to publicly share information (to be further defined in the Agreements) that does not disclose proprietary design or performance information, or information from which proprietary design or performance information can be derived.

8.6 Surveillance and Reporting

The Contracting Officer (CO) or the Contracting Officer's Representative (COR), as the CO's duly authorized representative, are authorized to perform inspection on behalf of the Government for the purpose of acceptance of all services to be provided.

The Participant must provide quarterly technical/schedule/risk progress reports to the CO, the COR, and the FAA Technical Point of Contact (POC) on the 15th (or first working day following) after the end of each calendar quarter covering the activities of the previous three months. The technical/schedule progress report must include technical accomplishments for the reporting period, accomplishments expected in the reporting period, descriptions of and reasons for any schedule changes, a high-level program schedule, and updates on any program risks.

The Participant must provide monthly financial reports, submitted to the COR and FAA Technical POC on the 15th (or the first working day following) of each calendar month, covering finances for the previous month. The monthly financial report must include company expenditures, FAA expenditures, and number of labor hours for that month and cumulatively, mapped to the CLEEN IV work elements. Copies of monthly invoices must be included in the

monthly financial report and explanations of any inconsistencies between invoices and the monthly financial reports must be provided.

At its discretion, the FAA will conduct site visits and attend technology demonstration tests. Every two weeks, technical, financial and schedule progress must be reviewed orally via teleconference with the Participant to assess progress toward completing the requirements set forth above. Presentation materials from these teleconferences are deliverable under the Agreement.

The Participant must conduct design review meetings for all major design milestones under the Agreement, including conceptual, preliminary, detailed, system, and critical design reviews, as applicable. These reviews should include FAA program office and other Government technical experts selected by the FAA program office. Materials from these reviews are deliverable under the Agreement.

The Participant must participate in a Government-led consortium, as directed by the FAA. The consortium will meet semi-annually. The basic function of the consortium meeting will be a forum for the presentation of the results of ongoing work. At these meetings, consortium members will present the progress and results of ongoing work in both government-only forums and public forums. FAA does not expect participants to share proprietary information with other consortium members. The Participant must provide access to the presentations with the third-party independent entity designated to conduct the benefits assessments and simulations. The Participant must provide all presentation material to the FAA Task Manager two weeks prior to the consortium meeting (10 working days). FAA will only require information for CLEEN-funded projects to be shared within the consortium. A Participant may choose to share information and results of other relevant research outside of CLEEN (subject to any applicable contractual and/or legal restrictions) but would not be required to do so. Subcontractors will not be required to participate in the consortium as long as the results of the subcontractors' work is shared with the consortium members, *e.g.*, by the Participant.

Inspection, review or the anticipation of acceptance/approval of an agreement item in the course of its preparation must not be construed as assurance of acceptance of the finished product.

All technical deliverables must have a 30 calendar-day review period, unless otherwise agreed to by both parties, wherein the Government has 30 calendar days to review and comment and the Participant has 30 days after receipt of comments from the Government to incorporate and address the comments in the deliverable. Deliverables are considered accepted if no comments are received 30 calendar days after submission.

The parties understand that data may be required from time to time to meet either Company or Agency reporting requirements. Upon mutual agreement, each Party will support the other in these data requests to the extent that the data is releasable, subject to applicable law and FAA policy.

8.7 Deliverable Schedule

8.7.1 Schedule

Document Title	Due Date	Format	Recipient
Task Plan	Initial plan with proposal. Revised final plan 30 days after task initiation	Electronic submission via secure FAA SharePoint site (KSN)	COR and Task Manager
Risk Assessment	30 days after Award		
Technical/Schedule/Risk Progress Reports	Quarterly		
Bi-weekly teleconference progress review materials	Every 2 weeks		
Monthly Financial Report	Monthly		
Design Review Agenda	5 working days prior to any conceptual, preliminary, or detailed design review		
Design Review Package	By the day of any conceptual, preliminary, or detailed design review		
Test and Instrumentation Plan	Determined on a project specific basis		
Informal Test Reports	Determined on a project specific basis		
Final Test Reports	Determined on a project specific basis		
CLEEN Consortium Presentation – Government-only Limited Rights Session	10 working days prior to the start day of each CLEEN Consortium Meeting, held semi-annually		
CLEEN Consortium Presentation – Public Session	10 working days prior to the start day of each CLEEN Consortium Meeting, held semi-annually		
Final Report(s) – Government-only Limited Rights Version (including both limited rights data and unlimited rights data)	Determined on a project specific basis	Meeting	
Final Report(s) – Publicly Releasable Version	Determined on a project specific basis		
Final Oral Briefing	Determined on a project specific basis	Meeting	

8.7.2 Data Item Descriptions

Document Title	Description
Task Plan	Captures planned work activities including any negotiated changes to Technical Proposal post-award. The Participant must maintain and control the Task Plan during the execution of this agreement
Risk Assessment	Risk Assessments must be prepared to identify possible high risk areas and appropriate plans for risk mitigation. Risk assessments also define critical program paths for each risk element and assess the risk of achieving technical objectives, goals, and schedules within budget. Plans and associated costs to provide risk reduction through alternate or backup approaches, if such action is required, must be included. Risk assessment methodology, trade-offs, and impact of risk reductions (alternate or backup approaches) on goals, schedules, and costs must also be included.
Technical/Schedule/Risk Progress Reports	The technical/schedule progress report must include technical accomplishments for the reporting period, accomplishments expected in the reporting period, descriptions of and reasons for any schedule changes, a high-level program schedule, and updates on any program risks.
Bi-weekly teleconference progress review materials	The presentation materials from bi-weekly progress review teleconferences, including presentation slides on technical progress, schedule, and cost updates, must be deliverable.
Monthly Financial Report	The monthly financial report must include company expenditures, FAA expenditures, and number of labor hours for that month and cumulatively, mapped to the CLEEN IV work elements. Copies of monthly invoices must be included in the monthly financial report and explanations of any inconsistencies between invoices and the monthly financial reports must be provided.
Design Review Agenda	5 working days prior to any conceptual, preliminary, or detailed design review
Design Review Package	By the day of any conceptual, preliminary, or detailed design review. To include presentation material and any relevant supporting documents.
Test and Instrumentation Plan	Determined on a project specific basis.
Informal Test Reports	An informal test report, submitted, within (number of days will be determined on a project specific basis) days of test completion, to the FAA Task Manager for review and approval. The report must include information on the test objectives accomplished and data acquired.
Final Test Reports	A final test report, submitted within (number of days will be determined on a project specific basis) of test completion, to the FAA Task Manager for review and approval. This report must include information on the test objectives accomplished, data acquired, interpretation and value of the data, and an updated assessment of the anticipated technology benefit, in terms of reduction in noise, emissions and /or fuel burn/energy use.

CLEEN Consortium Presentation – Government-only Limited Rights Session	Oral report and presentation materials on technical, financial, and schedule progress, as well as risks, including limited rights data, 10 working days prior to each CLEEN IV Consortium, held semiannually.
CLEEN Consortium Presentation – Public Session	Oral report and presentation materials on technical, financial, and schedule progress suitable for public release, 10 working days prior to each CLEEN IV Consortium, held semi-annually. This must include updated technology benefits relative to the CLEEN IV goals. Public presentations are to be compliant with Section 508 of the Rehabilitation Act of 1973, as amended (29 U.S.C. § 794 (d)).
Final Report(s) –Government-only Limited Rights Version (including both limited rights data and unlimited rights data)	The final report must have two versions, a limited rights version and an unlimited rights version. The limited rights final report version will contain all relevant data (including both limited rights data and unlimited rights data), for the FAA.
Final Report(s) – Publicly Releasable Version	The final report must have two versions, a limited rights version and an unlimited rights version. The unlimited rights final report version must contain descriptions of key research activities and milestones completed, benefits of the technologies towards the CLEEN goals, next steps to transition the research into future product applications, as well as other unlimited rights data as outlined in the agreement. Publicly Releasable Final Reports must be compliant with Section 508 of the Rehabilitation Act of 1973, as amended (29 U.S.C. § 794 (d)).
Final Oral Briefing	Final oral briefing to be conducted in Washington, DC or at another location or virtually at the discretion of the FAA. Written versions of this briefing containing all the data presented at the final oral briefing must be delivered to the FAA. If the oral presentation discloses information that has not been provided in written form at the briefing, the written version must be revised by including such information and delivered within 30 days of the oral briefing.

The Participant must include cost share progress reports in the financial progress reports required. Monthly invoices for work performed must be submitted. Financial progress reports must include detail in which both Participant funded and government funded work are identified by work element.

8.8 Points of Contact

The following personnel are the designated points of contact for this SIR:

Contracting Officer (CO)

Federal Aviation Administration

Attn: Kelly Chappelle

800 Independence Ave SW, AAQ-460, Washington, DC 20591

Tel: (202) 267-4631

Email: kelly.chappelle@faa.gov

Contracting Officer's Representative (COR)

Federal Aviation Administration

Attn: Jennifer White

800 Independence Ave SW, APL/AEE-3B, Washington, DC 20591

Tel: (202) 267-3577

Email: jennifer.white@faa.gov

9 Respondent Proposal Submissions

9.1 Proposal Volumes and Due Dates

Respondents must assemble and identify proposal volumes in accordance with the table below.

Respondents must ensure that all proposal volume titles and associated content are compliant with the page limitations and with the number of copies, as shown in the tables below.

Proposal Volume	Due Date / Time
Executive Summary Executive Summary – Technical Summary Spreadsheet Volume I - Technical/Management Proposal Volume II - Pricing/Business Proposal Volume III - Agreement Feedback	Date – May 18, 2026 Time - NLT 12:00 P.M. EDT

Submission notes:

- The FAA will not accept any submission made by facsimile, telex, telegraph, or similar devices.
- Respondents assume full responsibility for ensuring that the Contracting Officer receives proposals no later than the date and time specified above. Proposals should be submitted to Kelly Chappelle, Contracting Officer at 9-AFN-AWA-AAQ460-Contracting-Team@faa.gov.
- The FAA will consider proposals received by the Government after the specified deadline late and will exclude these proposals from evaluation.
- The Respondent states that the following persons are authorized to negotiate on the Respondent's behalf with the FAA in connection with this offer:

Name: _____

Title: _____

Phone number: _____

Email: _____

9.2 Notification of Intent to Withdraw Proposal

A Respondent may withdraw its proposal from competition by sending written notice, signed by an authorized negotiator, to the Contracting Officer at any time prior to Award. Proposals may be withdrawn via e-mail, but not by facsimile or mail.

9.3 SIR Questions, Concerns, and Clarifications

- The personnel identified in Section 8.8 are the sole point of contact for this acquisition. All Respondent questions, concerns, and clarification requests must be made in writing and submitted to the FAA via email to the Contracting Officer at email: 9-AFN-AWA-AAQ460-Contracting-Team@faa.gov no later than **12:00 PM, EDT, April 13, 2026**.
- The FAA will compile questions received and post them for review by all Respondents;

however, the FAA will not post the author of each question. The FAA may post some questions prior to the FAA responding to them. Prior to submitting a question, the Respondents must review posted questions and FAA responses to ensure the question has not already been addressed. If a Respondent needs further clarification on a previously submitted question, they must refer to that question by its identification number.

- (c) Written inquiries received within the timeframe specified in (a) above, will be answered in writing and provided to all Respondents prior to the required date for submittal of the Respondent's proposal. The FAA may not answer any requests received outside of the above timeframe. Furthermore, Respondents must note that the proposal due date will not be extended by the FAA due to questions received after the timeframe mentioned above.

9.4 Expenses Related to Respondent Submissions

- (a) The FAA will not pay for the information solicited, nor reimburse the Respondents for any costs incurred in the preparation or submission of any response to this SIR or in making necessary studies or designs for the preparation thereof.
- (b) The FAA will not allow any pre-award costs on this Agreement. Pre-award costs are defined as any costs incurred before the start date of the federal award in anticipation of the Federal award. Any pre-award costs at the Respondent's risk. The FAA will not approve or pay any pre-award costs.

9.5 Disposition of Proposals

The FAA will not return Respondent's proposals and will either retain or destroy them.

9.6 False Statements

Proposals must set forth full, accurate, and complete information as required by this solicitation, including attachments. The penalty for making false statements in proposals is prescribed in 18 U.S.C. § 1001.

9.7 Communications with Respondents

The Government may communicate at any time with Respondents on an individual basis to explain or clarify particular aspects of the solicitation, to negotiate as appropriate the terms and conditions of the proposed agreement, or to seek additional information regarding Respondent's submissions. Communication with potential Respondents may take place throughout the evaluation process. The purpose of communication is to ensure there is mutual understanding between FAA and the Respondents on all aspects of the solicitation, including the Respondent's proposals. The FAA may consider information disclosed because of oral or written communication with a Respondent in the evaluation of a Respondent's proposal(s).

To ensure that Respondents fully understand the intent of the SIR (and FAA's needs stated therein), the FAA may hold one-on-one meetings with individual Respondents pertaining to their proposals. One-on-one communications may continue throughout the process, as required, at the FAA's discretion. Communications with one Respondent do not necessitate communications with another Respondent because communications will be Respondent-specific. Regardless of the varying levels of communications with individual Respondents, the CO will ensure such communications do not give any Respondent an unfair competitive advantage.

Respondents must provide in their submissions Points of Contact (POCs), telephone numbers, and e-mail addresses for this purpose.

9.8 Responsible Prospective Respondent

Notwithstanding the evaluation methodology outlined in this SIR, the CO will determine whether the Respondent is responsible prior to the award of any resultant Agreement. As a minimum, in order to be determined responsible, a prospective Respondent must have: adequate financial resources to perform the Agreement, or the ability to obtain those resources; have a satisfactory record of integrity and business ethics; have a satisfactory performance record; have the necessary organization, experience, accounting and operational controls; and/or, be otherwise qualified and eligible to receive an award under applicable laws and regulations.

9.9 Acquisition Milestones

The following acquisition milestones are for informational purposes only and are subject to change:

Milestone Goal	Date
SIR Issued	March 2026
Proposals Due	12:00 p.m. EDT, May 18, 2026
Award of Agreement (s)	September 2026 (est.)

9.10 Proposal Preparation Instructions

Proposals must be complete and conform to the instructions in this section; incomplete proposals may result in the exclusion of such proposals from further consideration. General statements that the Respondent understands the requirements of the work to be performed in this effort or simple rephrasing or restating of the FAA's requirements, will not be considered adequate and will be reflected in lower evaluation scores or may be cause for rejection of the proposal.

All proposals will be screened initially for completeness, accuracy, and timeliness. Proposals that do not meet all initial screening criteria will not be evaluated further. It is the Respondent's responsibility to ensure the completeness of the proposal. The evaluation of proposals will be conducted on the basis of the information contained in the written proposal. The FAA will not assume that a Respondent possesses any capabilities not specified in the written proposal.

Proposals submitted must be formatted in accordance with the instructions provided in this section. Vendors must furnish proposals in the versions required below:

- Executive Summary - Electronic: Adobe PDF and Excel
- Volume 1 – Technical/Management Proposal - Electronic: Adobe PDF
- Volume 2 – Pricing/Business Proposal - Electronic: Adobe PDF
- Volume 3 – Agreement Feedback - Electronic: Adobe PDF

Format is standard 8 ½ x 11 paper, minimum 1 inch margins, left, right, top, and bottom, single side printing only, with text font size no less than 12. Charts, Graphs, Tables, and pictures are limited to a minimum font size of 8. Page counts do not include table of contents, list of figures, glossary, or cover pages. Foldouts are counted as one page, however, excessive use of foldouts is discouraged. Attachments are included in page count. Elaborate presentation techniques, including color, are neither required nor desired.

Executive Summary - (No more than two (2) pages – not assessed)

The Executive Summary must provide an overview of the Offeror's CLEEN IV Technology proposal(s) to include a completed summary Excel file attachment. This Executive Summary will not be used in the evaluation of proposals, and is simply intended to be a useful reference.

9.10.1 Volume 1 – Technical/Management Proposal

The Respondent's technical/management proposal must not exceed a total of fifteen (15) pages for a proposal containing a single CLEEN IV Technology project. This page limitation may be expanded for each additional CLEEN IV Technology being proposed.

The page limit for Factor 1 is no more than three (3) pages. The page limit for Factor 2 is no more than seven (7) pages per each proposed CLEEN IV Technology Project. The page limit for Factor 3 is no more than five (5) pages per each proposed CLEEN IV Technology Project.

Example page limits with variations on the number of projects are as follows (note that there is not a limit on the number of projects within a proposal):

Factors	Example Page Counts with			
	1 Project / Technology	2 Projects / Technologies	3 Projects / Technologies	4 Projects / Technologies
Factor 1: Corporate Experience	3	3	3	3
Factor 2: CLEEN IV Technology Benefits	7	14	21	28
Factor 3: CLEEN IV Technology Maturation Plan	5	10	15	20
Total Page Count	15	27	39	51

Factor 1: Corporate Experience (No more than three (3) pages of the total 15 pages)

The Respondent must describe its corporate experience providing services that are of a similar scope and complexity to the services contemplated in this SIR. Examples of corporate experience may include prior CLEEN awards, independent research and development (IRAD) efforts or related commercial or Government contracts, grants, or agreements. The proposal

must provide at least three (3) recent and relevant projects. For the purposes of Factor 1, the FAA defines Corporate Experience information to be recent and relevant as:

- (a) **Recent:** Projects performed either with the Government or in the commercial sector, that are either completed within the last ten (10) years from the date the SIR is released or are ongoing as of the date of the Respondent's submission.
- (b) **Relevant:** Projects of a similar scope and complexity to the effort described in Section 8.4 of this SIR.

The proposal must include the following information at a minimum: project title, agency and/or commercial/company name, project start-end dates, project value (\$), client point of contact (POC) information, including e-mail address and telephone number, description of work performed, and a description of why the work is relevant to CLEEN IV.

Factor 2: CLEEN IV Technology Benefits (No more than seven (7) pages per each proposed CLEEN IV Technology Project)

The proposal must address each of the following three (3) subfactors for each CLEEN IV Technology Project being proposed. Subfactors 2.1, 2.2, and 2.3 are of equal importance.

Subfactor 2.1: Potential to Inform the Development of FAA Environmental Standards and Environmental Reviews

The proposal must address how the project will assist the FAA in developing more effective and timely environmental standards and conducting environmental reviews. This must include a description of what information will be provided to the FAA that will: 1) assist the FAA in setting appropriate levels for environmental standards, 2) assist the FAA in improving certification procedures from an environmental perspective where the existing regulations and policy do not address or are not adequate for a given technology, and/or 3) assist the FAA in conducting environmental reviews of rulemaking and other actions.

Subfactor 2.2: Potential Aircraft and Fleet-Wide Benefits

The proposal must address the potential fuel burn, noise, and emissions benefits of the proposed CLEEN IV Technology at an aircraft level (relative to the metrics described in Section 8.2) and for future aircraft fleets operating in the National Airspace System. This includes the applicability of the technology across a range of aircraft sizes and its application to new aircraft and engine type designs (including new production of existing aircraft and engine type designs and retrofit of existing aircraft and engines). The proposal must address interdependencies, *i.e.*, the effect that a proposed aircraft or engine technology aimed at achieving one of the CLEEN IV objectives could have on the other objectives, either positive or negative. The proposal also must address the fuel burn, noise, and emissions impacts throughout the operational lifecycle of the CLEEN IV Technology based on the expected introduction date and fleet integration rate.

Subfactor 2.3: Potential for Technology Transition into the Fleet

The proposal must address the timeline and plan for introduction of CLEEN IV technology into the fleet. The proposal must assess the current technology readiness level, projected timeline for achieving FAA type certification, and level of risk involved

in maturing, certifying, and commercializing the technology. This includes how the results of the CLEEN IV research and testing activities will be used in the next steps to advance the proposed candidate technologies into future product engines and/or airframes. The proposal should also address the business case for the technology, including the near-term production potential, the cost of incorporating the proposed technology into aircraft and engines, and projected lifecycle costs, including the costs associated with reliability and maintainability.

Factor 3: CLEEN IV Technology Maturation Plan (No more than five (5) pages per each proposed CLEEN IV Technology Project)

The proposal must address each of the following two (2) subfactors for each CLEEN IV Technology Project being proposed. Subfactor 3.1 is more important than Subfactor 3.2.

Subfactor 3.1: Technical Approach

The proposal must provide an estimate of the current maturity and anticipated development of each candidate technology in terms of Technology Readiness Level (TRL) and identify the basis for that TRL ranking. Candidate technologies must be matured and demonstrated at a TRL sufficient to provide information to the FAA that supports the CLEEN IV performance objectives.

The technical approach must also address the following, if applicable to its proposed CLEEN IV Technology:

- A plan to develop the CLEEN IV Technology through design; fabrication; and component, rig, and/or full scale ground/flight tests to mature the technology, as applicable. Projects must provide data to support the CLEEN IV performance objectives.
- A plan to assess and advance manufacturability, repairability, maintainability, affordability, and certifiability of the technology during the proposed work plan.

The proposal must provide a technical approach and analytical plan for achieving its CLEEN IV objectives. This plan must include the data to be used, the source of that data within the program's activities, and a description of the calculation methods that will use that data to determine the benefits achieved against all applicable CLEEN IV performance objectives.

Subfactor 3.2: Program Management

The Respondent must describe its management approach to perform the work, including a clear assessment of primary risks and a means to address them; facilities, techniques, or unique combinations of these which are integral factors for achieving the proposal's objectives; and qualifications, capabilities, and experience of the proposed Program Manager, Principal Investigator and principal staff critical in achieving the proposal objectives. The Respondent's Management Approach must address the following:

- Provide Organizational Structure of the Respondent's Team and rationale for the structure. The proposal must also include an organizational chart that details the lines of authority and responsibility.
- Provide Program Manager, Principal Investigator, and principal staff summary qualifications and their contributions to the successful completion of the proposed technical approach and capability.
- Provide Risk Management approach to identify, assess, and mitigate risks in performance of this Agreement.
- Provide Cost/Schedule approach, methodology, and mechanism for monitoring and controlling cost and schedule variances during performance.
- Identify Facilities and Resources to be used in performance for this Agreement. The Respondent must provide a rationale for why the identified facilities and resources will contribute to the successful performance.

9.10.2 Volume 2 – Pricing/Business Proposal

Volume 2 - Pricing/Business Proposal will consist of two sections:

- Section 1: Proposed FAA contribution
- Section 2: Proposed Respondent contribution

The Respondent's Pricing/Business Proposal is not page limited. At a minimum, the Pricing/Business Proposal must include the following information:

Section 1: Proposed FAA Contribution

The pricing/business proposal must include the following information in support of the proposed FAA contribution (Respondent's reimbursement):

- Authorized Individuals – The Respondent must provide the name, title, telephone number, and e-mail address for the individual designated as the central point of contact for this proposal.
- Terms and Conditions – The Respondent must state whether its proposal is in complete compliance with the terms and conditions of this Agreement.
- Assumptions – The Respondent must describe any assumptions used to develop the proposed prices.

The cost proposal must identify the Respondent's estimated costs down to the sub-elements of each of the Work Elements described in the SIR, as applicable.

In addition to the above, the pricing proposal must include the following:

- a. Direct Labor: The labor category, number of hours, basis of estimate (historical or other), direct labor rate per hour for each category, and totals for each category of the effort.
- b. Materials: An itemized listing of material requirements for the effort.
- c. Travel: Contemplated expenditures for travel that support the effort.
- d. Other Direct Costs: Costs associated with laboratory usage, computer usage,

- reproduction, etc.
- e. Costs for Consultants: The number of hours and total costs must be indicated as well as the nature and scope of the service furnished by any consultant.
- f. Cost for Subcontractors: A breakdown of the subcontract amount by cost element
- g. Indirect Labor Cost: Overhead and general administrative rates.
- h. Funding Profile: Since award(s) under this Agreement may be incrementally funded, the Respondent must include its desired funding profile indicating planned expenditures by calendar quarters as well as cost matching if applicable.
- i. Total Price: Total proposed price of all cost elements referenced above (a- h).

Section 2: Proposed Respondent Contribution (cost/contribution of the parties)

- The Respondent must identify the specific contributions that are proposed in meeting the 1:1 minimum cost share requirement of the CLEEN program. S I R Section 6 describes in detail the requirements for this cost share.
- For facilities proposed to be used by the Respondent as part of their proposal submission, the Respondent must:
 1. Identify the facilities needed, and the required period or duration of use for such facilities or property.
 2. Make arrangements with the appropriate facility manager as to how the facilities will be made available and whether or not such use will be made available on a continuous non-interference basis.
 3. Provide a full-cost estimate for the use of such facilities from the facility manager.
 4. Provide a letter of commitment from the facility manager that addresses items 1 through 3 above.

9.10.3 Volume 3 – Agreement Feedback

Volume 3 of the proposal must be a copy of the attached sample agreement, Attachment A, Cooperative Agreement Template, in Microsoft Word format (.docx), with tracked changes enabled and comments on any terms, conditions, or articles that the Respondent does not find agreeable, and their proposed modification to those terms. This is intended to provide a starting point for negotiations, should a proposal be selected for award.

10 Proposal Selection and Evaluation

10.1 Proposal Selection

The FAA will use its discretion in selecting proposal(s) for award that potentially provide the greatest benefit to the public and the Government in achieving the objectives of the CLEEN Program. The FAA will consider technical merit and the value of the proposal to the CLEEN Program; cost/contribution of the parties; and the capability of the Respondent to accomplish the objectives of CLEEN IV Technology Project(s) identified in its proposal. The FAA will consider the degree to which the full portfolio of awards under the CLEEN program addresses all performance objectives of the program and will consider this in its determination of the benefit to the Government. FAA reserves the right to make trade-off decisions between these considerations

and is not confined to making award to either the Respondent submitting the highest-rated technical proposal or the lowest-priced proposal, although FAA may award to either.

The FAA reserves the right to award an Agreement immediately following the evaluation of the initial proposal and may not undertake communications. Therefore, it is critical that the initial proposal be fully responsive to this SIR and that the proposal contains the Respondent's best terms.

The FAA reserves the right to limit further consideration of proposals to only those Respondents with the best likelihood of receiving an award. FAA reserves the right to make partial awards as a result of this solicitation. Additionally, the FAA reserves the right to make multiple awards, a single award, or no awards at all under this solicitation.

The FAA may select specific CLEEN IV Technology projects as part of a partial award. If technology project proposals are offered as a non-separable package of work, the Respondent should specify this in the proposal. In such a case, the FAA will not make partial awards on such proposals. The proposed scope of work may include manufacturing if it is required to develop and mature a specific aircraft technology. The FAA reserves the right to only partially fund any award(s) it issues as a result of this solicitation.

There is no restriction on the number of proposals that an organization may submit to this SIR or on the teaming arrangements for any one proposal. However, each proposal must be a separate, stand-alone, complete document for evaluation purposes. Government agencies and Federally Funded Research and Development Centers are not eligible to submit proposals under this SIR or participate as team members under proposals submitted by eligible entities.

10.1.1 Evaluation Factors Order of Importance

For the Selection determination:

- Under Volume 1, Factor 1, Corporate Experience, will be assessed as “Pass” or “Fail”. A proposal receiving a “Fail” assessment will not be eligible for award consideration.
- Under Volume 1, Factor 2, CLEEN IV Technology Benefits, is considered more important than Factor 3, CLEEN IV Technology Maturation Plan.
- The subfactors under Factor 2, CLEEN IV Technology Benefits, are considered of equal importance.
- Under Factor 3, CLEEN IV Technology Maturation Plan, Subfactor 3.1, Technical Approach, is considered more important than Subfactor 3.2, Program Management.
- Volume 1 is considered more important than Volume 2. If the FAA determines that a competing proposal demonstrates more benefit to the Government, then a higher cost proposal may be selected for award, where the value of the selected proposal is determined to support the higher cost.
- Volume 3 will be reviewed but will not be rated. However, the FAA will consider whether any significant proposed changes to the sample agreement by the Respondent would reduce the value of the proposal to the government.

10.2 Proposal Evaluation

10.2.1 Volume 1 – Technical/Management Proposal

Factor 1: Corporate Experience (Pass/Fail)

Corporate experience will be evaluated based on the relevance of the Respondent's experience related to the CLEEN IV work elements of similar scope and complexity to the services contemplated in this SIR. To receive a rating of "Pass", the Respondent must provide at least three (3) recent and relevant projects. For the purposes of Factor 1, the FAA defines Corporate Experience information to be recent and relevant as:

- (c) **Recent:** Projects performed either with the Government or in the commercial sector, that are either completed within the last ten (10) years from the date the SIR is released or are ongoing as of the date of the Respondent's submission.
- (d) **Relevant:** Projects of a similar scope and complexity to the effort described in Section 8.4 of this SIR.

The proposal must include the following information at a minimum: project title, agency and/or commercial/company name, project start-end dates, project value (\$), client point of contact (POC) information including e-mail address and telephone number, description of work performed, and a description of why the work is relevant to CLEEN IV.

Respondents that fail to provide at least three (3) recent and relevant projects in accordance with these requirements will receive a rating of "Fail" and will not be eligible for award. Also, the FAA reserves the right not to evaluate the technical proposal of any Respondent who received a "Fail" rating for this requirement.

Factor 2: CLEEN IV Technology Benefits (Adjectival Rating)

Each of the subfactors below will be assigned an adjectival rating, as defined in Section 10.3. These subfactors are considered of equal importance.

Subfactor 2.1: Potential to Inform the Development of FAA Environmental Standards and Environmental Reviews

The FAA will evaluate the degree to which the Respondent demonstrates a comprehensive and credible plan for providing information to the FAA that will assist the FAA in developing more effective and timely environmental standards and conducting environmental reviews, including a detailed description of the information that will be provided.

Subfactor 2.2: Potential Aircraft and Fleet-Wide Benefits

The FAA will evaluate the degree to which the Respondent demonstrates a comprehensive and credible plan to address the potential fuel burn, noise, and emissions benefits of the proposed CLEEN IV Technology at an aircraft level and for future aircraft fleets operating in the National Airspace System. This includes the applicability of the technology across a range of aircraft sizes and its application to new aircraft and engine

type designs (including new production of existing aircraft and engine type designs and retrofit of existing aircraft and engines).

The FAA will evaluate the extent to which the proposal addresses the fuel burn, noise, and emissions impacts throughout the operational lifecycle of the proposed CLEEN IV Technology based on expected introduction date and fleet integration rate. The FAA will also evaluate how the proposal accounts for interdependencies across performance objectives.

Subfactor 2.3: Potential for Technology Transition into the Fleet

The FAA will evaluate the degree to which the Respondent demonstrates a comprehensive and credible plan to address the timeline and plan for introduction of CLEEN IV technology into the fleet. The FAA will evaluate the degree to which the proposal assesses the current technology readiness level, projected timeline for achieving FAA type certification, and level of risk involved in maturing, certifying, and commercializing the technology. This includes how the results of the CLEEN IV research and testing activities will be used in next steps to advance the proposed candidate technologies into future product engines and/or airframes. The FAA will evaluate how the proposal addresses the business case for the technology, including the near-term production potential, the cost of incorporating the proposed technology into aircraft and engines, and projected lifecycle costs, including the costs associated with reliability and maintainability.

Factor 3: CLEEN IV Technology Maturation (Adjectival Rating)

Each of the subfactors below will be assigned an adjectival rating, as defined in Section 10.3. Subfactor 3.1 is considered to be more important than Subfactor 3.2.

Subfactor 3.1: Technical Approach

The FAA will evaluate the proposal's estimate of the current maturity and anticipated development of each candidate technology in terms of Technology Readiness Level (TRL) and the basis for that TRL ranking.

The FAA will evaluate the degree to which the Respondent demonstrates a comprehensive and credible technical approach and analytical plan in achieving its applicable CLEEN IV performance objectives.

The FAA will evaluate the proposed technical approach for the following (if applicable to its proposed CLEEN IV Technology):

- A plan to develop the CLEEN IV Technology through design; fabrication; and component, rig, and/or full scale ground/flight tests to mature the technology, as applicable. Projects must provide data to support the CLEEN IV performance objectives.
- A plan to assess and advance manufacturability, repairability, maintainability,

affordability and certifiability of the technology during the proposed work plan.

The FAA will evaluate the proposed approach for data utilization, the source of that data within the program's activities, and description of the calculation methods that will use that data to determine the benefits achieved against all applicable CLEEN IV performance objectives.

Subfactor 3.2: Program Management

The FAA will evaluate the proposed plan and methodology to ensure effective technology transfer and knowledge dissemination for the purposes of promoting transition of the CLEEN IV Technology. This includes how the results of the CLEEN IV research and testing activities will be used in next steps to advance the proposed candidate technologies into future product engines and/or airframes.

The FAA will evaluate the Respondent's proposed management approach to perform the work, including a clear assessment of primary risks and a means to address them; facilities, techniques, or unique combination of these which are integral factors for achieving the proposal's objectives; and qualifications, capabilities, and experience of the proposed Program Manager, Principal Investigator and principal staff critical in achieving the proposal objectives. All program management items are of equal importance under Factor 3. The FAA will evaluate the degree to which the proposal provides a comprehensive and credible approach to the following program management items:

- Organizational Structure of the Respondent's Team and its rationale for the structure. The proposed organizational chart that details the lines of authority and responsibility.
- Program Manager, Principal Investigator, and principal staff summary qualifications and their contributions to the successful completion of the proposed technical approach and capability.
- Risk Management approach to identify, assess, and mitigate risks in performance of this Agreement.
- Cost/Schedule approach, methodology, and mechanism for monitoring and controlling cost and schedule variances during performance.
- Facilities and Resources identified to ensure successful performance for this Agreement. A rationale for why the identified facilities and resources will contribute to the successful performance.

10.2.2 Volume 2 – Pricing/Business Proposal Evaluation

The FAA will perform a price analysis of the proposal(s). The price evaluators will review the proposed elements of price and basis of estimate(s) for fairness and reasonableness. Interested Respondents should provide enough information in their pricing/business proposal for the FAA to be able to do so. The FAA reserves the right to do a cost analysis or a realism analysis, at the FAA's discretion, if it deems that necessary.

The FAA will assess the suitability and value of the Respondent's proposed cost sharing

contributions. The Respondent must match Federal funds on at least a 1:1 basis. The FAA will assess the suitability of the proposed cost sharing contributions based on the following criteria:

- (a) Determine whether each element of the proposed cost share contribution is an acceptable cost share item as defined in the CLEEN IV work elements identified in Section 8.4 and as specified in Section 6 Funding.
- (b) For the proposed cost share elements that are determined to be acceptable, determine whether the proposed value of each cost share element is reasonable.
- (c) Calculate the sum of the acceptable cost share elements to determine whether the total cost share meets the 1:1 minimum cost share requirement established for the CLEEN IV Program.

The FAA will assess the reasonableness of the full cost estimate provided for the use of any facilities included in the Respondent's proposal. The FAA may contact the identified negotiator(s) in order to validate the Respondent's full cost estimate provided in the Respondent's proposal.

10.2.3 Volume 3 - Agreement Feedback

Volume 3 will be reviewed for compliance with the requirements of Section 9.10.3, but will not be rated. However, the FAA will consider whether any significant proposed changes to the sample agreement by the Respondent would reduce the value of the proposal to the government.

10.3 Factor 2, CLEEN IV Technology Benefits, and Factor 3, CLEEN IV Technology Maturation Plan Adjectival Ratings

The following adjectival ratings will be used to evaluate Volume I, Factor 2, CLEEN IV Technology Benefits, Factor 3, CLEEN IV Technology Maturation Plan, and the Factor 2 and 3 subfactors.

An adjectival rating will be assigned to each of the three (3) Factor 2 subfactors and each of the two (2) Factor 3 subfactors. The adjectival ratings of each corresponding subfactor under Factors 2 and 3 will be aggregated into one overall adjectival rating for that factor.

Rating	Definition
Excellent	Comprehensive and complete response. The impact of strengths far outweighs the combined impact of weaknesses and risks.
Good	Substantial response. The impact of strengths outweighs the combined impact of weaknesses and risks.
Satisfactory	Moderate response. The impact of strengths is approximately equal to the combined impact of weaknesses and risks.
Marginal	Marginal response. The combined impact of weaknesses and risks outweighs the impact of strengths.

Unsatisfactory	Inadequate response. The combined impact of weaknesses and risks far outweighs the impact of strengths.
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Below are the definitions of terms used in the adjectival ratings.

Term	Definition
Strength	An aspect of a proposal that would positively impact performance of the resulting cooperative agreement, exceed the minimum requirements, or otherwise benefit the Government.
Weakness	An aspect of a proposal that would negatively impact performance of the resulting cooperative agreement, fail to meet the minimum requirements, or otherwise harm the Government.
Risk	An aspect of the Respondent's proposal that presents an uncertainty as to the ability of the Respondent to successfully perform the resulting cooperative agreement or the proposed approach.

11 Period of Offer

Respondent's proposals must be binding for at least 180 calendar days from the date of the final submission. Proposals may offer more than 180 days.

12 Additional Clauses and Provisions

The clauses and provisions below will be incorporated into CLEEN IV agreements.

12.1 Officials Not to Benefit

No member of or delegate to Congress, or resident commissioner is permitted to any share or part of the Agreement, or to any benefit arising from it. However, this clause does not apply to the Agreement to the extent that the Agreement is made with a corporation for the corporation's general benefit.

12.2 Disclosure Regarding Payments to Influence Certain Federal Transactions

(a) Definitions.

- (1) "The Act," as used in this clause, means section 1352, title 31, United States Code.
- (2) "Agency," as used in this clause, means executive agency, within the meaning of 5 U.S.C. 101, 102, and 104(I), and any wholly owned Government corporation within the meaning of 31 U.S.C. 9101.
- (3) "Covered Federal action," as used in this clause, means any of the following Federal actions:
 - (i) The awarding of any Federal contract.
 - (ii) The making of any Federal grant.
 - (iii) The making of any Federal loan.

- (iv) The entering into of any cooperative agreement.
 - (v) The extension, continuation, renewal, amendment, or modification of any Federal contract, grant, loan, or cooperative agreement.
- (4) "Indian tribe" and "tribal organization," as used in this clause, have the meaning provided in section 4 of the Indian Self-Determination and Education Assistance Act (25 U.S.C. 5304) and include Alaskan Natives.
- (5) "Influencing or attempting to influence," as used in this clause, means making, with the intent to influence, any communication to or appearance before an officer or employee of any agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with any covered Federal action.
- (6) "Local government," as used in this clause, means a unit of government in a State and, if chartered, established, or otherwise recognized by a State for the performance of a governmental duty, including a local public authority, a special district, an intrastate district, a council of governments, a sponsor group representative organization, and any other instrumentality of a local government.
- (7) "Officer or employee of an agency," as used in this clause, includes the following individuals who are employed by an agency:
- (i) An individual who is appointed to a position in the Government under title 5, United States Code, including a position under a temporary appointment.
 - (ii) A member of the uniformed services, as defined in subsection 101(3), title 37, United States Code.
 - (iii) A special Government employee, as defined in section 202, title 18, United States Code.
 - (iv) An individual who is a member of a Federal advisory committee, as defined by the Federal Advisory Committee Act, title 5, United States Code, appendix 2.
- (8) "Person," as used in this clause, means an individual, corporation, company, association, authority, firm, partnership, society, State, and local government, regardless of whether such entity is operated for profit, or not for profit. This term excludes an Indian tribe, tribal organization, or any other Indian organization with respect to expenditures specifically permitted by other Federal law.
- (9) "Reasonable compensation," as used in this clause, means, with respect to a regularly employed officer or employee of any person, compensation that is consistent with the normal compensation for such officer or employee for work that is not furnished to, not funded by, or not furnished in cooperation with the Federal Government.
- (10) "Reasonable payment," as used in this clause, means, with respect to professional and other technical services, a payment in an amount that is consistent with the amount normally paid for such services in the private sector.
- (11) "Recipient," as used in this clause, includes the Participant and all subcontractors. This term excludes an Indian tribe, tribal organization, or any other Indian organization with respect to expenditures specifically permitted by other Federal law.
- (12) "Regularly employed," as used in this clause, means, with respect to an officer or employee of a person requesting or receiving a Federal contract, an officer or employee who is employed by such person for at least 130 working days within 1 year immediately preceding the date of the submission that initiates agency consideration of such person for receipt of such contract. An officer or employee who is employed by such person for less than 130 working days within 1 year immediately preceding the date of the

submission that initiates agency consideration of such person must be considered to be regularly employed as soon as the officer or employee is employed by such person for 130 working days.

(13) "State," as used in this clause, means a State of the United States, the District of Columbia, the Commonwealth of Puerto Rico, a territory or possession of the United States, an agency or instrumentality of a State, and multi-State, regional, or interstate entity having governmental duties and powers.

(b) Prohibitions. The participant, by signing its offer, hereby certifies to the best of his or her knowledge and belief that:

(1) No Federal appropriated funds have been paid or will be paid to any person for influencing or attempting to influence an officer or employee of any agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress on his or her behalf in connection with the awarding of any Federal contract, the making of any Federal grant, the making of any Federal loan, the entering into of any cooperative agreement, and the extension, continuation, renewal, amendment or modification of any Federal contract, grant, loan, or cooperative agreement;

(2) If any funds other than Federal appropriated funds (including profit or fee received under a covered Federal action) have been paid, or will be paid, to any person for influencing or attempting to influence an officer or employee of any agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress on his or her behalf in connection with the screening information request (SIR), the respondent must complete and submit, with its offer, OMB Standard Form LLL, Disclosure of Lobbying Activities, to the Contracting Officer; and

(3) The participant will include the language of this clause in all subcontract awards at any tier and require that all recipients of subcontract awards in excess of \$150,000 must disclose accordingly.

(4) This certification and disclosure is a prerequisite for making or entering into the Agreement imposed by the Act. Any person who makes a prohibited expenditure or fails to file or amend a disclosure form, must be subject to a civil penalty of not less than \$10,000 and not more than \$100,000, for each such failure.

(c) The prohibitions of the Act do not apply under the following conditions:

(1) Agency and legislative liaison by its own employees.

(i) The prohibition on the use of appropriated funds, in subparagraph (b)(1) of this clause, does not apply in the case of a payment of reasonable compensation made to an officer or employee of a person requesting or receiving a covered Federal action if the payment is for agency and legislative liaison activities not directly related to a covered Federal action.

(ii) For purposes of subdivision (c)(1)(i) of this clause, providing any information specifically requested by an agency or Congress is permitted at any time.

(iii) The following agency and legislative liaison activities are permitted at any time where they are not related to a specific solicitation for any covered Federal action:

(A) Discussing with an agency the qualities and characteristics (including individual demonstrations) of the person's products or services, conditions or terms of sale, and service capabilities.

(B) Technical discussions and other activities regarding the application or adaptation of the person's products or services for an agency's use.

(iv) The following agency and legislative liaison activities are permitted where they are prior to the Screening Information Request (SIR) of any covered Federal action:

(A) Providing any information not specifically requested but necessary for an agency to make an informed decision about initiation of a covered Federal action;

(B) Technical discussions regarding the preparation of an unsolicited proposal prior to its official submission; and

(C) Capability presentations by persons seeking awards from an agency pursuant to the provisions of a law authorizing such actions;

(2) Professional and technical services.

(i) The prohibition on the use of appropriated funds, in subparagraph (b)(1) of this clause, does not apply in the case of:

(A) A payment of reasonable compensation made to an officer or employee of a person requesting or receiving a covered Federal action or an extension, continuation, renewal, amendment, or modification of a covered Federal action, if payment is for professional or technical services rendered directly in the preparation, submission, or negotiation of submittal/offer or application for that Federal action or for meeting requirements imposed by or pursuant to law as a condition for receiving that Federal action.

(B) Any reasonable payment to a person, other than an officer or employee of a person requesting or receiving a covered Federal action or an extension, continuation, renewal, amendment, or modification of a covered Federal action if the payment is for professional or technical services rendered directly in the preparation, submission, or negotiation of any submittal/offer or application for that Federal action or for meeting requirements imposed by or pursuant to law as a condition for receiving that Federal action. Persons other than officers or employees of a person requesting or receiving a covered Federal action include consultants and trade associations.

(ii) For purposes of subdivision (c)(2)(i) of this clause, 'professional and technical services' must be limited to advice and analysis directly applying any professional or technical discipline. For example, drafting of a legal document accompanying a submittal/offer by a lawyer is allowable. Similarly, technical advice provided by an engineer on the performance or operational capability of a piece of equipment rendered directly in the negotiation of a contract is allowable. However, communications with the intent to influence made by a professional (such as a licensed lawyer) or a technical person (such as a licensed accountant) are not allowable under this section unless they provide advice and analysis directly applying their professional or technical expertise and unless the advice or analysis

is rendered directly and solely in the preparation, submission or negotiation of a covered Federal action. Thus, for example, communications with the intent to influence made by a lawyer that do not provide legal advice or analysis directly and solely related to the legal aspects of his or her client's submittal/offer, but generally advocate one proposal over another are not allowable under this section because the lawyer is not providing professional legal services. Similarly, communications with the intent to influence made by an engineer providing an engineering analysis prior to the preparation or submission of a submittal/offer are not allowable under this section since the engineer is providing technical services but not directly in the preparation, submission or negotiation of a covered Federal action.

(iii) Requirements imposed by or pursuant to law as a condition for receiving a covered Federal award include those required by law or regulation and any other requirements in the actual award documents.

(iv) The reporting requirements herein must not apply with respect to payments of reasonable compensation made to regularly employed officers or employees of a person.

(3) Only those communications and services expressly authorized by paragraphs (c)(1) and (2) of this clause are permitted under this clause.

(d) Disclosure.

(1) If the Participant, who requests or receives from an agency a Federal contract, has made or has agreed to make any payment using non-appropriated funds (to include profits from any Covered Federal action), to any person for the purpose of influencing or attempting to influence an officer or employee of any Agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress on his or her behalf in connection with a Covered Federal action, the Participant must file with that agency a disclosure form, OMB Standard Form LLL, Disclosure of Lobbying Activities..

(2) The Participant must file a disclosure form at the end of each calendar quarter in which there occurs any event that materially affects the accuracy of the information contained in any disclosure form previously filed by such person under subparagraph (d)(1) of this clause. An event that materially affects the accuracy of the information reported includes:

- (i) A cumulative increase of \$25,000 or more in the amount paid or expected to be paid for influencing or attempting to influence a covered Federal action; or
- (ii) A change in the person(s) or individual(s) influencing or attempting to influence a covered Federal action; or
- (iii) A change in the officer(s), employee(s), or Member(s) contacted to influence or attempt to influence a covered Federal action.

(3) The Participant must require the certification, and if required, a disclosure form by any person who requests or receives any subcontractor exceeding \$150,000 under the Federal contract.

(4) All subcontractor disclosure forms must be forwarded from tier to tier until received by the prime Participant. The prime Participant must submit all disclosures to the Contracting Officer at the end of the calendar quarter in which the disclosure form is submitted by the subcontractor.

(e) Agreement. The Participant agrees not to make any payment prohibited by this clause.

(f) Penalties.

(1) Any person who makes an expenditure prohibited under paragraph (b) of this clause or fails to file or amend the disclosure form to be filed or amended by paragraph (b) must be subject to civil penalties as provided for by 31 U.S.C. 1352. An imposition of a civil penalty does not prevent the Government from seeking any other remedy that may be applicable.

(2) Participants may rely without liability on the representations made by their subcontractors in the certification and in the disclosure form.

(g) Cost allowability. Nothing in this clause makes allowable or reasonable any costs which would otherwise be unallowable or unreasonable. Conversely, costs made specifically unallowable by the requirements in this clause will not be made allowable under any other provision.

12.3 Limitation of Cost

Notwithstanding any other provision herein, and unless expressly agreed in writing, the FAA's total cash contribution, will in no event exceed [\$ amount]. Furthermore, the FAA's obligation under this agreement is contingent upon the availability of appropriated funds. Except as expressly stated in this Agreement, the FAA assumes no liability or obligation in connection with [Company Name]'s technologies. Participant must make required notifications and follow all direction in accordance with the below provisions.

(a) The parties estimate that performance of the Agreement, exclusive of any fee, will not cost the FAA more than the FAA's share of the estimated cost specified in the Agreement. The Participant agrees to use its best efforts to perform the work specified in the Agreement and all obligations under the Agreement within the estimated cost, which includes both the FAA's and the Participant's share of the cost.

(b) The Participant must notify the Contracting Officer (CO) in writing whenever it has reason to believe that-

(1) The costs the Participant expects to incur under the Agreement in the next 60 days, when added to all costs previously incurred, will exceed 75 percent of the estimated cost specified in the Agreement; or

(2) The total cost for the performance of the Agreement, exclusive of any fee, will be either greater or substantially less than had been previously estimated.

(c) As part of the notification, the Participant must provide the CO a revised estimate of the total cost of performing the Agreement.

(d) Except as required by other provisions of the Agreement, specifically citing and stated to be an exception to this clause-

(1) The FAA is not obligated to reimburse the Participant for costs incurred in excess of the estimated cost to the FAA specified in the Agreement and:

(2) The Participant is not obligated to continue performance under this Agreement (including actions under the "Termination" clause of the Agreement) or otherwise incur costs in excess of the estimated cost specified in the Agreement, until the CO:

(i) Notifies the Participant in writing that the estimated cost has been increased and

(ii) Provides a revised estimated total cost of performing the Agreement.

The increase will be allocated in accordance with the formula specified in the Agreement.

(e) No notice, communication, or representation in any form other than that specified in subparagraph (d)(2) above, or from any person other than the CO, will affect this agreement's estimated cost to the FAA. In the absence of the specified notice, the FAA is not obligated to reimburse the Participant for any costs in excess of the estimated cost to the FAA specified in the Agreement, whether those excess costs were incurred during the course of the agreement or as a result of termination.

(f) If the estimated cost specified in the Agreement is increased, any costs the Participant incurs before the increase that are in excess of the previously estimated cost will be allowable to the same extent as if incurred afterward, unless the CO issues a termination or other notice directing that the increase is solely to cover termination or other specified expenses.

(g) Change orders will not be considered an authorization to exceed the estimated cost to the FAA specified in the Agreement, unless they contain a statement increasing the estimated cost.

(h) If this agreement is terminated or the estimated cost is not increased, the FAA and the Participant will negotiate an equitable distribution of all property produced or purchased under the agreement, based upon the share of costs incurred by each.

12.4 Reimbursement for Travel and Subsistence

(a) Travel and subsistence are authorized for work performed under the Agreement. The Participant will be reimbursed for allocable, allowable, and reasonable travel expenses only at the per diem rates specified in applicable Federal travel regulations and in accordance with 2 CFR § 200.475, Travel Costs.

(b) The Participant must conduct a cost analysis prior to the start of travel to determine the most cost-effective means of travel. Alternate airports must be used where available and within a reasonable commuting distance if it will result in lower costs.

(c) The Participant must summarize the travel (date and place of the expenses, purpose of the trip, name of person(s) of trip, and title or relationship to Participant) and submit the information in accordance with the invoicing instructions in the Agreement. The Participant must submit its travel

expense report to the Contracting Officer at the same time the Participant submits its invoice for payment.

(d) Travel will be reimbursed in accordance with the above and the Participant's provisional G&A rate applicable to the time the travel occurred.

12.5 Patent Rights – Ownership by the Participant

(a) Definitions. As used in this clause -

"Invention" means any invention or discovery that is or may be patentable or otherwise protectable under title 35 of the United States Code, or any variety of plant that is or may be protectable under the Plant Variety Protection Act (7 U.S.C. 2321, *et seq.*).

"Made" means -

(1) When used in relation to any invention other than a plant variety, the conception or first actual reduction to practice of the invention; or

(2) When used in relation to a plant variety, that the Participant has at least tentatively determined that the variety has been reproduced with recognized characteristics.

"Nonprofit organization" means a university or other institution of higher education or an organization of the type described in section 501(c)(3) of the Internal Revenue Code of 1954 (26 U.S.C. 501(c)) and exempt from taxation under section 501(a) of the Internal Revenue Code (26 U.S.C. 501(a)) or any nonprofit scientific or educational organization qualified under a State nonprofit organization statute.

"Practical application" means to manufacture, in the case of a composition of product; to practice, in the case of a process or method; or to operate, in the case of a machine or system; and, in each case, under such conditions as to establish that the invention is being utilized and that its benefits are, to the extent permitted by law or Government regulations, available to the public on reasonable terms.

"Subject invention" means any invention of the Participant made in the performance of work under the Agreement.

(b) Participant's rights

(1) Ownership.

The Participant may retain ownership of each subject invention throughout the world in accordance with the provisions of this clause.

(2) License.

(i) The Participant must retain a nonexclusive royalty-free license throughout the world in each subject invention to which the Government obtains title, unless the Participant fails to disclose the invention within the times specified in paragraph (c) of this clause. The Participant's license extends to any domestic subsidiaries and affiliates, within the corporate structure of which the Participant is a part, and includes the right to grant sublicenses to the extent the Participant was legally obligated to do so at agreement award. The license is transferable only with the written approval of the agency, except when transferred to the successor of that part

of the Participant's business to which the invention pertains.

(ii) The Participant's license may be revoked or modified by the agency to the extent necessary to achieve expeditious practical application of subject invention in a particular country, in accordance with procedures at 37 CFR Part 404.

(c) Participant's obligations.

(1) The Participant must disclose in writing each subject invention to the Contracting Officer within 2 months after the inventor discloses it in writing to Participant personnel responsible for patent matters. The disclosure must identify the inventor(s) and the agreement under which the invention was made. It must be sufficiently complete in technical detail to convey a clear understanding of the subject invention. The disclosure must also identify any publication, on sale (*i.e.*, sale or offer for sale), or public use of the subject invention or whether a manuscript describing the invention has been submitted for publication and, if so, whether it has been accepted for publication. In addition, after disclosure to the agency, the Participant must promptly notify the Contracting Officer of the acceptance of any manuscript describing the invention for publication, and any on sale or public use.

(2) The Participant must elect in writing whether or not to retain ownership of any subject invention by notifying the Contracting Officer within 2 years of disclosure to the agency. However, in any case where publication, on sale, or public use has initiated the 1-year statutory period during which valid patent protection can be obtained in the United States, the period for election of title may be shortened by the agency to a date that is no more than 60 days prior to the end of the statutory period.

(3) The Participant must file either a provisional or nonprovisional patent application or a Plant Variety Protection Application on an elected subject invention within 1 year after election. However, in any case where a publication, sale, or public use has initiated the 1-year statutory period during which valid patent protection can be obtained in the United States, the Participant must file the application prior to the end of the statutory period. If the Participant files a provisional application, it must file a nonprovisional application within 10 months of the filing of the provisional application. The Participant must file patent applications in additional countries or international patent offices within either 10 months of the first filed patent application (whether provisional or nonprovisional) or 6 months from the date permission is granted by the Commissioner of Patents to file foreign patent applications where such filing has been prohibited by a Secrecy Order.

(4) The Participant may request extensions of the time for disclosure, election, or filing under subparagraphs (c) (1), (c)(2), and (c)(3) of this clause.

(5) The Participant may establish and maintain active and effective procedures to ensure that subject inventions are promptly identified and timely disclosed, and must submit a description of the procedures to the Contracting Officer so that the Contracting Officer may evaluate and determine their effectiveness.

(d) Government rights –

(1) Ownership. The Participant must assign to the agency, upon written request, title to any

subject invention-

(i) If the Participant fails to disclose or elect ownership to the subject invention within the times specified in paragraph (c) of this clause, or elects not to retain ownership; provided, that the agency may request title only within 60 days after learning of the Participant's failure to disclose or elect within the specified times.

(ii) In those countries in which the Participant fails to file patent applications within the times specified in paragraph (c) of this clause; provided, however, that if the Participant has filed a patent application in a country after the times specified in paragraph (c) of this clause, but prior to its receipt of the written request of the agency, the Participant must continue to retain ownership in that country.

(iii) In any country in which the Participant decides not to continue the prosecution of any application for, to pay the maintenance fees on, or defend in reexamination or opposition proceeding on, a patent on a subject invention.

(2) License. If the Participant retains ownership of any subject invention, the Government must have a nonexclusive, nontransferable, irrevocable, paid-up license to practice, or have practiced for or on its behalf, the subject invention throughout the world.

(e) Participant action to protect the Government's interest.

(1) The Participant must execute or to have executed and promptly deliver to the agency all instruments necessary to -

(i) Establish or confirm the rights the Government has throughout the world in those subject inventions in which the Participant elects to retain ownership; and

(ii) Assign title to the agency when requested under paragraph (d) of this clause and to enable the Government to obtain patent protection and plant variety protection for that subject invention in any country.

(2) The Participant must require, by written agreement, its employees, other than clerical and nontechnical employees, to disclose promptly in writing to personnel identified as responsible for the administration of patent matters and in the Participant's format, each subject invention in order that the Participant can comply with the disclosure provisions of paragraph (c) of this clause, and to execute all papers necessary to file patent applications on subject inventions and to establish the Government's rights in the subject inventions. The disclosure format should require, as a minimum, the information required by subparagraph (c)(1) of this clause. The Participant must instruct such employees, through employee agreements or other suitable educational programs, as to the importance of reporting inventions in sufficient time to permit the filing of patent applications prior to U.S. or foreign statutory bars.

(3) The Participant must notify the Contracting Officer of any decisions not to file a nonprovisional patent application, continue the prosecution of a patent application, pay maintenance fees, or defend in a reexamination or opposition proceeding on a patent, in any country, not less than 30 days before the expiration of the response or filing period required by the relevant patent office.

(4) The Participant must include, within the specification of any United States nonprovisional patent or plant variety protection application and any patent or plant variety

protection certificate issuing thereon covering a subject invention, the following statement, "The invention was made with Government support under (identify the agreement) awarded by (identify the agency). The Government has certain rights in the invention."

(f) Reporting on utilization of subject inventions.

The Participant must submit, on request, periodic reports no more frequently than annually on the utilization of a subject invention or on efforts at obtaining utilization of the subject invention that are being made by the Participant or its licensees or assignees. The reports must include information regarding the status of development, date of first commercial sale or use, gross royalties received by the Participant, and such other data and information as the agency may reasonably specify. The Participant also must provide additional reports as may be requested by the agency in connection with any march-in proceeding undertaken by the agency in accordance with paragraph (h) of this clause. The Participant must also mark any utilization report as confidential/proprietary to help prevent inadvertent release outside the Government. As required by 35 U.S.C. 202(c)(5), the agency will not disclose such information to persons outside the Government without the Participant's permission.

(g) Preference for United States industry.

Notwithstanding any other provision of this clause, neither the Participant nor any assignee must grant to any person the exclusive right to use or sell any subject invention in the United States unless such person agrees that any products embodying the subject invention or produced through the use of the subject invention will be manufactured substantially in the United States. However, in individual cases, the requirement for an agreement may be waived by the agency upon a showing by the Participant or its assignee that reasonable but unsuccessful efforts have been made to grant licenses on similar terms to potential licensees that would be likely to manufacture substantially in the United States, or that under the circumstances domestic manufacture is not commercially feasible.

(h) March-in rights.

The Participant acknowledges that, with respect to any subject invention in which it has retained ownership, the agency has the right to require licensing pursuant to 35 U.S.C. 203 and 210(c), and in accordance with the procedures in 37 CFR 401.6 and any supplemental regulations of the agency in effect on the date of agreement award.

(i) Special provisions for agreements with nonprofit organizations.

If the Participant is a nonprofit organization, it must -

- (1) Not assign rights to a subject invention in the United States without the written approval of the agency, except where an assignment is made to an organization that has as one of its primary functions the management of inventions, provided, that the assignee will be subject to the same provisions as the Participant;
- (2) Share royalties collected on a subject invention with the inventor, including Federal employee co-inventors (but through their agency if the agency deems it appropriate) when the subject invention is assigned in accordance with 35 U.S.C. 202(e) and 37 CFR 401.10;

(3) Use the balance of any royalties or income earned by the Participant with respect to subject inventions, after payment of expenses (including payments to inventors) incidental to the administration of subject inventions for the support of scientific research or education; and,

(4) Make efforts that are reasonable under the circumstances to attract licensees of subject inventions that are small business concerns, and give a preference to a small business concern when licensing a subject invention if the Participant determines that the small business concern has a plan or proposal for marketing the invention which, if executed, is equally as likely to bring the invention to practical application as any plans or proposals from applicants that are not small business concerns; provided, that the Participant is also satisfied that the small business concern has the capability and resources to carry out its plan or proposal. The decision whether to give a preference in any specific case will be at the discretion of the Participant.

(5) Allow the Secretary of Commerce to review the Participant's licensing program and decisions regarding small business applicants, and the negotiate changes to its licensing policies, procedures, or practices with the Secretary of Commerce when the Secretary's review discloses that the Participant could take reasonable steps to more effectively implement the requirements of paragraph (i)(4) of this clause.

(j) Communications. [Complete according to agency instructions]

(k) Subcontracts.

(1) The Participant must include the substance of this clause, including this paragraph (k), in all subcontracts for experimental, developmental, or research work to be performed by a small business concern or nonprofit organization.

(2) The Participant must include in all other subcontracts for experimental, developmental, or research work the substance of the patent rights clause required to adequately protect the Government's interests consistent with section 3.5 of the FAA Acquisition Management System.

(3) At all tiers, the patent rights clause must be modified to identify the parties as follows: references to the Government are not changed, and the subcontractor has all rights and obligations of the Participant in the clause. The Participant must not, as part of the consideration for awarding the subcontract, obtain rights in the subcontractor's subject inventions.

(4) In subcontracts, at any tier, the agency, subcontractor, and the Participant agree that the mutual obligations of the parties created by this clause constitute an agreement between the subcontractor and the agency with respect to the matters covered by the clause; provided, however, that nothing in this paragraph is intended to confer any jurisdiction under the FAA disputes resolution process, or any board or judicial proceeding, in connection with proceedings under paragraph (h) of this clause, dealing with march-in rights.

12.6 Rights in Data – General

(a) Definitions. As used in this clause -

"Commercial Computer Software" means -

- (1) Computer software, that is of a type customarily used by the general public or by non-governmental entities for purposes other than governmental purposes, and -
 - (i) Has been sold, leased, or licensed to the general public; or,
 - (ii) Has been offered for sale, lease, or license to the general public;
- (2) Any computer software that evolved from computer software described in paragraph (1) of this definition through advances in technology or performance and that is not yet available in the commercial marketplace, but will be available in the commercial marketplace in time to satisfy the delivery requirements under a Government solicitation;
- (3) Any item that would satisfy a criterion expressed in paragraphs (1) or (2) of this definition, but for -
 - (i) Modifications of a type customarily available in the commercial marketplace; or
 - (ii) Minor modifications of a type not customarily available in the commercial marketplace made to meet Federal Government requirements. "Minor modifications" means modifications that do not significantly change the purpose of the computer software. Factors to be considered in determining whether a modification is minor include the value and size of the modification and the comparative value and size of the final product. Dollar values and percentages may be used as guideposts, but are not conclusive evidence that a modification is minor;
- (4) Any combination of items meeting the requirements of paragraphs (1), (2), (3), of this definition that are of a type customarily combined and sold in combination to the general public; or
- (5) Restricted Computer Software, if the procuring agency determines the restricted computer software is sold in substantial quantities, on a competitive basis, to multiple State and local governments.

"Computer database" or "database means" a collection of recorded information in a form capable of, and for the purpose of, being stored in, processed, and operated on by a computer. The term does not include computer software.

"Computer software" means -

- (i) Computer programs that comprise a series of instructions, rules ,routines, or statements, regardless of the media in which they are recorded, that allow or cause a computer to perform a specific operation or series of operation, and
- (ii) Recorded information comprising source code listings, design details, algorithms, processes, flow charts, formulas, and related material that would enable the computer program to be produced, created or compiled.

Computer software does not include computer databases or computer software documentation.

"Computer software documentation" means owner's manuals, user's manuals, installation instructions, operating instructions, and other similar items, regardless of storage medium, that explains the capabilities of the computer software or provide instructions for using the software.

"Data" means recorded information, regardless of form or the media on which it may be recorded. The term includes technical data and computer software. The term does not include information

incidental to agreement administration, such as financial, administrative, cost or pricing, or management information.

"Form, fit, and function data" means data relating to items, components, or processes that are sufficient to enable physical and functional interchangeability, and data identifying source, size, configuration, mating, and attachment characteristics, functional characteristics, and performance requirements. For computer software it means data identifying source, functional characteristics, and performance requirements but specifically excludes the source code, algorithms, processes, formulae, and flow charts of the software.

"Limited rights" means the rights of the Government in limited rights data as set forth in the Limited Rights Notice of subparagraph (g)(2) if included in this clause.

"Limited rights data" means data, other than computer software, that embody trade secrets or are commercial or financial and confidential or privileged, to the extent that such data pertain to items, components, or processes developed at private expense, including minor modifications.

"Restricted computer software," means computer software developed at private expense and that is a trade secret, is commercial or financial and is confidential or privileged, or is copyrighted computer software, including minor modifications of the computer software.

"Restricted rights" means the rights of the Government in restricted computer software, as set forth in a Restricted Rights Notice of subparagraph (g) if included in this clause, or as otherwise may be provided in a collateral agreement incorporated in and made part of this agreement, including minor modifications of such computer software.

"Technical data" means recorded information (regardless of the form or method of the recording) of a scientific or technical nature (including computer databases and computer software documentation). This term does not include computer software or financial, administrative, cost or pricing, or management data or other information incidental to agreement administration. The term includes recorded information of a scientific or technical nature that is included in computer databases.

"Unlimited rights" means the rights of the Government to use, disclose, reproduce, prepare derivative works, distribute copies to the public, and perform publicly and display publicly, in any manner and for any purpose, and to have or permit others to do so.

(b) Allocations of rights.

(1) Except as provided in paragraph (c) of this clause, the Government must have unlimited rights in-

- (i) Data first produced in the performance of this agreement;
- (ii) Form, fit, and function data delivered under this agreement;
- (iii) Data delivered under this agreement (except for restricted computer software) that constitute manuals or instructional and training material for installation, operation, or routine maintenance and repair of items, components, or processes delivered or furnished for use under this agreement; and
- (iv) All other data delivered under this agreement unless provided otherwise for limited rights data or restricted computer software in accordance with paragraph (g) of this clause.

(2) The Participant must have the right to -

- (i) Assert copyright in data first produced in performance of this agreement to the extent provided in paragraph (c)(1) of this clause;
- (ii) Use, release to others, reproduce, distribute, or publish any data first produced or specifically used by the Participant in the performance of this agreement, unless provided otherwise in paragraph (d) of this clause;
- (iii) Substantiate use of, add or correct limited rights, restricted rights, or copyright notices and to take other appropriate action, in accordance with paragraphs (e) and (f) of this clause; and
- (iv) Protect from unauthorized disclosure and use those data which are limited rights data or restricted computer software to the extent provided in paragraph (g) of this clause;

(c) Copyright.

(1) Data first produced in the performance of this agreement. Except as otherwise specifically provided in this agreement, the Participant may assert copyright in any data first produced in the performance of this agreement. When asserting copyright, the Participant must affix the applicable copyright notice of 17 U.S.C. 401 or 402 and an acknowledgment of Government sponsorship (including agreement number), to the data when such data are delivered to the Government, as well as when the data are published or deposited for registration as a published work in the U.S. Copyright Office. For data other than computer software, the Participant grants to the Government, and others acting on its behalf, a paid-up, nonexclusive, irrevocable, worldwide license for all such data to reproduce, prepare derivative works, distribute copies to the public, and perform publicly and display publicly, by or on behalf of the Government. For computer software, the Participant grants to the Government and others acting on its behalf, a paid up, nonexclusive, irrevocable, worldwide license for all such computer software to reproduce, prepare derivative works, and perform publicly and display publicly (but not to distribute copies to the public), by or on behalf of the Government.

- (i) Unless provided otherwise in paragraph (d) of this clause, the Participant may, without prior approval of the Contracting Officer, assert copyright in scientific and technical articles based on or containing data first produced in the performance of this agreement and published in academic, technical or professional journals, symposia proceedings or similar works. The prior, express written permission of the Contracting Officer is required to assert copyright subsisting in all other data first produced in the performance of this agreement.
- (ii) When authorized to assert copyright to the data, the Participant must affix the applicable copyright notices of 17 U.S.C. 401 or 402 and acknowledgment of Government sponsorship (including agreement number).
- (iii) For data other than computer software, the Participant grants to the Government, and others acting on its behalf, a paid-up, nonexclusive, irrevocable worldwide license in such copyrighted data to reproduce, prepare derivative works, distribute copies to the public, and perform publicly and display publicly, by or on

behalf of the Government. For computer software, the Participant grants to the Government and others acting in its behalf, a paid-up nonexclusive, irrevocable worldwide license in such copyrighted computer software to reproduce, prepare derivative works, and perform publicly and display publicly (but not to distribute copies to the public) by or on behalf of the Government.

(2) Data not first produced in the performance of this agreement. The Participant must not, without prior written permission of the Contracting Officer, incorporate in data delivered under this agreement any data not first produced in the performance of this agreement unless the Participant

(i) Identifies such data, and

(ii) Grants to the Government, or acquires on its behalf, a license of the same scope as set forth in subparagraph (c)(1) of this clause or, if such data are computer software, the Government must acquire a copyright license as set forth in subparagraph (g)(4) of this clause (if included in this agreement) or as otherwise may be provided in a collateral agreement incorporated in or made part of this agreement.

(3) Removal of copyright notices. The Government will not remove any authorized copyright notices placed on data pursuant to this paragraph (c), and will include such notices on all reproductions of the data.

(d) Release, publication and use of data. The Participant must have the right to use, release to others, reproduce, distribute, or publish any data first produced or specifically used by the Participant in the performance of this agreement, except

(1) As prohibited by Federal law or regulation (*e.g.*, export control or national security laws or regulations);

(2) As expressly set forth in this agreement; or,

(3) If the Participant receives or is given access to data necessary for the performance of this agreement that contain restrictive markings, the Participant must treat the data in accordance with such markings unless specifically authorized in writing by the Contracting Officer.

(e) Unauthorized marking of data.

(1) Notwithstanding any other provisions of this agreement concerning inspection or acceptance, if any data delivered under this agreement are marked with the notices specified in paragraph (g)(3) or (g)(4) if included in this clause, and use of the notices is not authorized by this clause, or if the data bears any other restrictive or limiting markings not authorized by this agreement, the Contracting Officer may at any time either return the data to the Participant, or cancel or ignore the markings. While the FAA is not subject to the requirements of 41 U.S.C. § 4703, nor to the procedures of the Contract Disputes Act at 41 U.S.C. §§ 7101-7109, the following procedures must apply prior to canceling or ignoring the markings.

(i) The Contracting Officer must make written inquiry to the Participant affording

the Participant 60 days from receipt of the inquiry to provide written justification to substantiate the propriety of the markings;

(ii) If the Participant fails to respond or fails to provide written justification to substantiate the propriety of the markings within the 60-day period (or a longer time approved in writing by the Contracting Officer for good cause shown), the Government must have the right to cancel or ignore the markings at any time after said period and the data will no longer be made subject to any disclosure prohibitions.

(iii) If the Participant provides written justification to substantiate the propriety of the markings within the period set in paragraph (e)(1)(i) of this clause, the Contracting Officer will consider such written justification and determine whether or not the markings are to be canceled or ignored. If the Contracting Officer determines that the markings are authorized, the Participant will be so notified in writing. If the Contracting Officer determines, with concurrence of the Chief of the Contracting Office [COCO], that the markings are not authorized, the Contracting Officer must provide a written determination to the Participant. If the Participant disagrees with the Contracting Officer determination, the Participant may seek resolution of the dispute under Article VII. DISPUTE RESOLUTION.

The Government must continue to abide by the markings under this paragraph (e)(1)(iii) until final resolution of the matter under Article VII. DISPUTE RESOLUTION (in which instance the Government will thereafter have the right to cancel or ignore the markings at any time and the data will no longer be made subject to any disclosure prohibitions).

(2) The time limits in the procedures set forth in paragraph (e)(1) of this clause may be modified in accordance with agency regulations implementing the Freedom of Information Act (5 U.S.C. 552) if necessary to respond to a request thereunder.

(f) Omitted or incorrect markings.

(1) Data delivered to the Government without any restrictive markings must be deemed to have been furnished with unlimited rights. The Government is not liable for the disclosure, use, or reproduction of such data.

(2) If the unmarked data has not been disclosed without restriction outside the Government, the Participant may request, within 6 months (or a longer time approved by the Contracting Officer for good cause shown) after delivery of such data, permission to have authorized notices placed on the data at the Participant's expense. The Contracting Officer may agree to do so if the Participant-

(i) Identifies the data to which the omitted notice is to be applied;

(ii) Demonstrates that the omission of the notice was inadvertent; and,

(iii) Establishes that the use of the proposed notice is authorized;

(iv) Acknowledges that the Government has no liability for the disclosure, use, or reproduction of any such data made prior to the addition of the notice or resulting from the omission of the notice.

- (3) If data has been marked with an incorrect notice, the Contracting Officer may -
- (i) Permit correction of the notice at the Participant's expense if the Participant identifies the data and demonstrates that the correct notice is authorized, or
 - (ii) Correct any incorrect notices.

(g) Protection of limited rights data and restricted computer software.

(1) The Participant may withhold from delivery qualified limited rights data or restricted computer software that are not identified in paragraphs (b)(1)(i), (ii), and (iii) of this clause. As a condition to this withholding, the Participant must -

- (i) Identify the data being withheld; and
- (ii) Furnish form, fit, and function data instead.

(2) Limited rights data that are formatted as a computer data base for delivery to the Government are to be treated as limited rights data and not restricted computer software.

(3) Notwithstanding subparagraph (g)(1) of this clause, the agreement may identify and specify the delivery of limited rights data, or the Contracting Officer may require by written request the delivery of limited rights data that has been withheld or would otherwise be entitled to be withheld. If delivery of such data is required, the Participant must affix the following "Limited Rights Notice" to the data and the Government will treat the data, subject to the provisions of paragraphs (e) and (f) of this clause, in accordance with the notice:

LIMITED RIGHTS NOTICE (January 2009)

- (a) These data are submitted with limited rights under Government Agreement No. XXXXXX (and subcontract N/A, if appropriate). These data may be reproduced and used by the Government with the express limitation that they will not, without written permission of the Participant, be used for purposes of manufacture nor disclosed outside the Government; except that the Government may disclose these data outside the Government for the following purposes, if any; provided that the Government makes such disclosure subject to prohibition against further use and disclosure:

Further study of the data by a Government contractor, educational or non-profit institution under provisions of a properly executed non-disclosure agreement for independent assessment; provisions of this non-disclosure agreement would extend to works and products derived from original limited rights data.

- (b) This Notice must be marked on any reproduction of these data, in whole or in part.

(h) Subcontracting. The Participant must obtain from its subcontractors all data and rights therein necessary to fulfill the Participant's obligations to the Government under this agreement. If a subcontractor refuses to accept terms affording the Government those rights, the Participant must promptly notify the Contracting Officer of the refusal and must not proceed with subcontract

award without authorization in writing from the Contracting Officer.

(i) Relationship to patents or other rights. Nothing contained in this clause must imply a license to the Government under any patent or be construed as affecting the scope of any license or other right otherwise granted to the Government.

12.7 Confidentiality of Data and Information

(a) In performance of this agreement, the Participant and any of its subcontractors, may need access to and use various data and information in the possession of the Government. This data and information may have been obtained under conditions which restrict the Government's right to use and disclose this data and information or which may be adverse to the interests of the Government or other parties if it is disseminated or used in a capacity other than in performance of this agreement. Therefore, the Participant and its subcontractors agree to abide by any restrictive use conditions on such data and not to: (1) knowingly disclose such data and information to others without written authorization from the Contracting Officer, unless it is already publicly available; or (2) use for any purpose other than the performance of this agreement any data or information which bears a restrictive marking or legend which the Participant has gained access to through the performance of this agreement, or information that should be marked according to FAA Order 1600.75 "Protecting Sensitive Unclassified Information (SUI)". For the sole purpose of this clause, "information" means any communication or representation of knowledge such as facts, data, or opinions in any medium or form, including textual, numerical, graphic, cartographic, narrative or visual form.

(b) In the event the work required to be performed under this agreement requires access to proprietary data and information of other companies, the Participant must obtain agreement from such other companies for such use unless such data are provided or made available to the Participant by the Government. Two copies of such company-to-company agreements must be furnished promptly to the Contracting Officer for information only. These agreements must prescribe the scope of authorized use and disclosure of the proprietary data and information as well as any other terms and conditions to be agreed upon between the parties thereto. It is agreed by the Participant that any such data or information, whether obtained by the Participant pursuant to the aforesaid agreement or from the Government, must be protected from unauthorized use by or unauthorized disclosure to any individual, corporation, or organization so long as it remains proprietary.

(c) The Participant agrees to conduct formal training to make employees aware of the requirement to maintain confidentiality of data and information as required above. The Participant must obtain from each employee in connection with this agreement a signed Non-Disclosure Agreement. This agreement must provide that the employee will not, during employment or anytime thereafter, disclose or use for current or future benefit of any party any of the data (to include any form of Sensitive Unclassified Information (SUI) described in FAA Order 1600.75) or information not publically available received in connection with the work under the agreement.

(d) The Participant agrees to hold the Government harmless and indemnify the Government as to any cost/loss resulting from the unauthorized use or disclosure of third party data or software by the Participant, its employees, subcontractors, or agents.

(e) The Participant agrees to include the substance of this clause in all subcontracts awarded under this agreement. The Contracting Officer will consider case-by-case exceptions to this requirement for individual subcontracts in the event that: (1) the Participant considers this clause to be inappropriate and unnecessary in the case of a particular subcontract; (2) the Participant provides a written statement affirming absolute unwillingness of a subcontractor to perform, absent some relief from the substance of this prohibition and the reason why; (3) use of an alternate subcontract source would unreasonably detract from the quality of effort; and (4) the Participant provides the Contracting Officer timely written advance notice of these and any other extenuating circumstances.

(f) Except as the Contracting Officer specifically authorizes in writing, upon completion of all work under this agreement, the Participant must return all such data and information described above obtained from the Government, including all copies, modifications, adaptations, or combinations thereof, to the Contracting Officer. Data obtained from another company must be disposed of in accordance with the Participant's agreement with that company, or if the agreement makes no provision for disposition, must be returned to that company. The Participant must further certify in writing to the CO that all copies, modifications, adaptations, or combinations of such data or information which cannot reasonably be returned to the Contracting Officer (or to the appropriate company), have been deleted from the Participant's (and any subcontractor's) records and destroyed. The FAA reserves the right to audit the deletion. The FAA must provide notice of the audit 10 calendar days prior to the audit.

(g) These restrictions do not limit the Participant's (or subcontractor's) right to use and disclose any data and information obtained from another source without restriction.

12.8 Subcontracts

(a) Subcontract, as used in this clause, includes but is not limited to purchase orders, and changes and modifications to purchase orders. The Participant shall notify the Contracting Officer reasonably in advance of entering into any subcontract if:

- (1) The proposed subcontract is of the cost-reimbursement, time-and-materials, or labor-hour type;
- (2) The proposed subcontract is fixed-price and exceeds either \$250,000 or 5 percent of the total estimated cost of this agreement;
- (3) The proposed subcontract has experimental, developmental, or research work as one of its purposes; or
- (4) This agreement is not a facilities contract and the proposed subcontract provides for the fabrication, purchase, rental, installation, or other acquisition of special test equipment valued in excess of \$25,000 or of any items of facilities.

(b)

(1) In the case of a proposed subcontract that

- (i) is of the cost-reimbursement, time-and-materials, or labor-hour type and is estimated to exceed \$25,000, including any fee,
- (ii) is proposed to exceed \$250,000, or
- (iii) is one of a number of subcontracts with a single subcontractor, under this agreement, for the same or related supplies or services that, in the aggregate, are expected to exceed \$250,000, the advance notification required by paragraph (a) above shall include the information specified in subparagraph (2) below.

(2)

- (i) A description of the supplies or services to be subcontracted.
- (ii) Identification of the type of subcontract to be used.
- (iii) Identification of the proposed subcontractor and an explanation of why and how the proposed subcontractor was selected, including the competition obtained.
- (iv) The proposed subcontract price and the Participant's cost or price analysis.
- (v) The subcontractor's current, complete, and accurate cost or pricing data and Certificate of Current Cost or Pricing Data, if required by other agreement provisions.
- (vi) The subcontractor's Disclosure Statement or Certificate relating to Cost Accounting Standards when such data are required by other provisions of this agreement.
- (vii) A negotiation memorandum reflecting-
 - (A) The principal elements of the subcontract price negotiations;
 - (B) The most significant considerations controlling establishment of initial or revised prices;
 - (C) The reason cost or pricing data were or were not required;
 - (D) The extent, if any, to which the Participant did not rely on the subcontractor's cost or pricing data in determining the price objective and in negotiating the final price;
 - (E) The extent to which it was recognized in the negotiation that the subcontractor's cost or pricing data were not accurate, complete, or current; the action taken by the Participant and the subcontractor; and the effect of any such defective data on the total price negotiated;
 - (F) The reasons for any significant difference between the Participant's price objective and the price negotiated; and
 - (G) A complete explanation of the incentive fee or profit plan when incentives are used. The explanation shall identify each critical performance element, management decisions used to quantify each incentive element, reasons for the incentives, and a summary of all trade-off possibilities considered.

(c) The Participant shall obtain the Contracting Officer's written consent before placing any subcontract for which advance notification is required under paragraph (a) above. However, the Contracting Officer may ratify in writing any such subcontract. Ratification shall constitute the consent of the Contracting Officer.

(d) If the Participant has an approved purchasing system and the subcontract is within the scope of such approval, the Participant may enter into the subcontracts described in subparagraphs (a)(1) and (a)(2) of this clause without the consent of the Contracting Officer.

(e) Even if the Participant's purchasing system has been approved, the Participant shall obtain the Contracting Officer's written consent before placing subcontracts identified below:

(f) Unless the consent or approval specifically provides otherwise, neither consent by the Contracting Officer to any subcontract nor approval of the Participant's purchasing system shall constitute a determination:

- (1) of the acceptability of any subcontract terms or conditions,
- (2) of the allowability of any cost under this agreement, or
- (3) to relieve the Participant of any responsibility for performing this agreement.

(g) No subcontract placed under this contract shall provide for payment on a cost-plus-a-percentage-of-cost basis.

(h) The Participant shall give the Contracting Officer immediate written notice of any action or suit filed and prompt notice of any claim made against the Participant by any subcontractor or vendor that, in the opinion of the Participant, may result in litigation related in any way to this agreement, with respect to which the Participant may be entitled to reimbursement from the Government.

(i)

(1)Reserved.

(2) Additionally, the Participant shall include in each cost- reimbursement subcontract under this agreement a requirement that the subcontractor insert the substance of the appropriate modified subparagraph referred to in subparagraph (1) above in each lower tier price redetermination or incentive price revision subcontract under that subcontract.

(j) To facilitate small business participation in subcontracting, the Participant agrees to provide payments on subcontracts under this agreement that are fixed-price subcontracts with small business concerns in conformity with the standards for customary payments, as in effect on the

date of this agreement. The Participant further agrees that the need for such financing payments will not be considered a handicap or adverse factor in the award of subcontracts.

(k) The Government reserves the right to review the Participant's purchasing system.

13 Attachments

Attachment A – Cooperative Agreement Template

Attachment B – CLEEN IV Technical Summary Spreadsheet (xlsx file)